

ANALOG & DIGITAL CIRCUITS

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About me

- Education:
 - BSc in Computer Science, MSc in Software Engineering, PhD in Computer Networks – TUCN
- I teach
 - Analog and digital lectures + labs
 - Computer networks (Ro) lectures + labs
 - Computer networks design labs
 - Wireless Technologies and Mobile Devices labs
 - CISCO Academy
- Research interests:
 - Computer networks, Computer networks design & protocols, Quality of Service in heterogeneous networks, NGNs, SDNs, sensor networks and their applications
 - IT&C applied in other domains: environment, health, societal change

Structure

- Lectures: 2 Hours/Week
- Labs: 2 Hours/Week
- MS Teams: CAN_ADC 2025_2026 class
 - Enroll using your TUCN account;
 - Team code: **520c93n**
 - Any problems/questions -> email me (Bogdan.lancu@cs.utcluj.ro)
- Moodle page: *course materials; labs materials + weekly assignments*
 - ***<https://moodle.cs.utcluj.ro/course/view.php?id=767>***
 - Self-enrolments key: **C@n_key2025**
 - Enrollment info – also, on MS Teams class

Module overview

- Content of Lectures

1. Introduction; Basic presentation for: pulse signal, RC circuits, pulse transformer
2. Switching regime of the semiconductor devices
3. Analog Circuits: Operational Amplifiers, Power supply sources, Oscillators
4. Digital Circuits: Parameters of the integrated logic circuits
5. Families of integrated logic circuits
6. Digital circuits for bus design
7. Trigger Schmitt circuits
8. Flip-flops
9. Semiconductor Memories
10. Converters: ADC & DAC
11. Microcontrollers (sample 80C51 family, Arduino family)

- Attendance

- Following regulations (*for bonuses: quizzes – TBD*)

Lab overview

- Structure
 - Simulation of circuit behavior
 - Circuits parameters in static & dynamic regime / in time & frequency
 - Applications & Design
 - Theoretical problems & discussions
 - Computer science view
- Attendance
 - Mandatory

Bibliography

- A. Peculea, B. Iancu, V. Dadarlat, S. Buzura, "Analog and digital Circuits. Practical applications", U.T. Press Cluj-Napoca, 2020
- A. Sedra, K.C. Smith, T.C. Carusone, Vincent Gaudet, "Microelectronic Circuits", Oxford University Press, 8th edition, 2019.
- P. Horowitz, W. Hill, "The Art of Electronics 3rd Edition", Cambridge University Press, 2015
- W. Oskay, E. Schlaepfer, "Open Circuits: The Inner Beauty of Electronic Components", No Starch Press, 2022
- V. Dădârlat, A. Peculea, "Circuite Analogice si Numerice", UTPress, 2006
- V. Dădârlat, "Circuite și dispozitive numerice", Ed. Mediamira, Cluj-Napoca, 1999
- Moodle page

Assessment

- **Lab** test + activity during labs (uploading your solutions, participating in discussions, explanations, etc)
- Written **Exam** (theory, problems)
 - 40% laboratory + 60% final exam
 - Grading constraints: minimum of 5 for lab and exam/exams
- Module Credits: 4

Questions

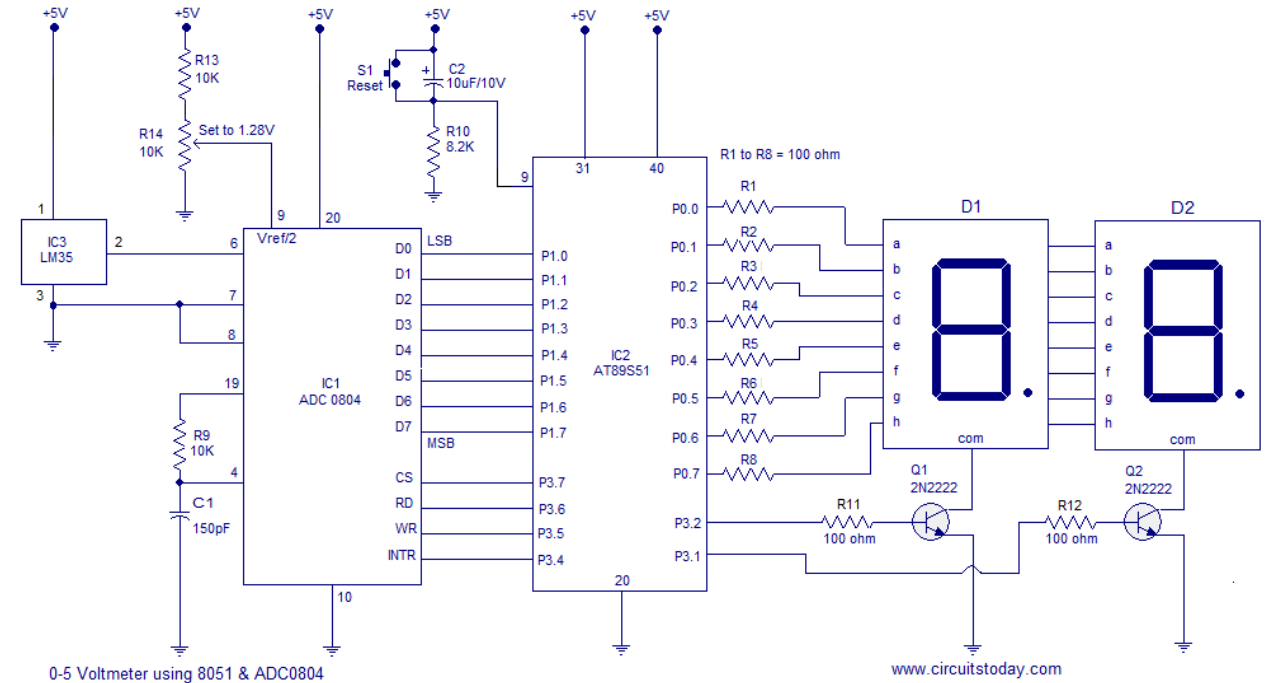
- What are:
 - Analog signals
 - Digital signals
- What are Analog and Digital Circuits?
- Examples and applications?

ADC in different domains

Examples

Sensors

- Digital thermometer
 - Temperature sensor – analog;
 - provides current or voltage output proportional to the absolute temperature
 - Analog-to-digital converter (ADC)
 - Converts the analog output voltage to a 8 bit digital value
 - Microcontroller
 - Display

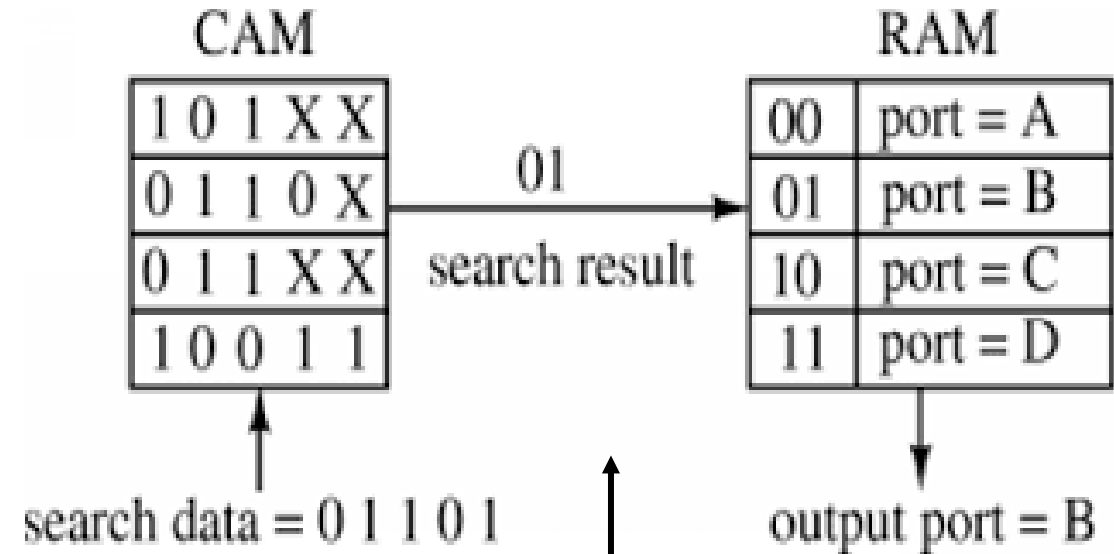


<https://www.circuitstoday.com/thermometer-using-8051>

Memories in network devices

- Read-only memory (ROM)
- Random-access memory (RAM)

- Content-addressable memory (CAM)
 - a.k.a associative memory or associative storage
 - find the address from the data (opposite to classic memory types)
 - more expensive and holds much less data
 - fast table lookups in **network routers and switches**, cache memory
 - SRAM-based memory + additional logic comparison circuitry delivers the results in one clock cycle



Integrated circuit chips

- Moore's law:
 - the number of integrated circuit elements per unit area will be doubled in two or two and half years.

MOSFET scaling (process nodes)

10 μm – 1971
6 μm – 1974
3 μm – 1977
1.5 μm – 1981
1 μm – 1984
800 nm – 1987
600 nm – 1990
350 nm – 1993
250 nm – 1996
180 nm – 1999
130 nm – 2001
90 nm – 2003
65 nm – 2005
45 nm – 2007
32 nm – 2009
22 nm – 2012
14 nm – 2014
10 nm – 2016
7 nm – 2018
5 nm – 2020
Future
3 nm ~ 2022
2 nm ~ 2024

Examples for CPUs

Intel 14th gen

- 10nm
- Transistors: ?

AMD Ryzen 5th gen

- 4nm and 3nm
- Transistors: 100+ billion for AMD Epyc

Apple M4 chip

- Transistors: 28 billion
- 3nm

Taiwan Semiconductor Manufacturing Company, Limited (TSMC)

Modeling complex biological systems

- Example: Modeling of cardiac cells
 - “The **rhythmic**, pump-like function of the heart is driven by muscle **contractions**, which are triggered by **electrical signals**”
 - “**Electrical waves** then travel along a prescribed path, exciting cells in the main chambers of the heart (atria and ventricles) and assuring **synchronous contractions**”

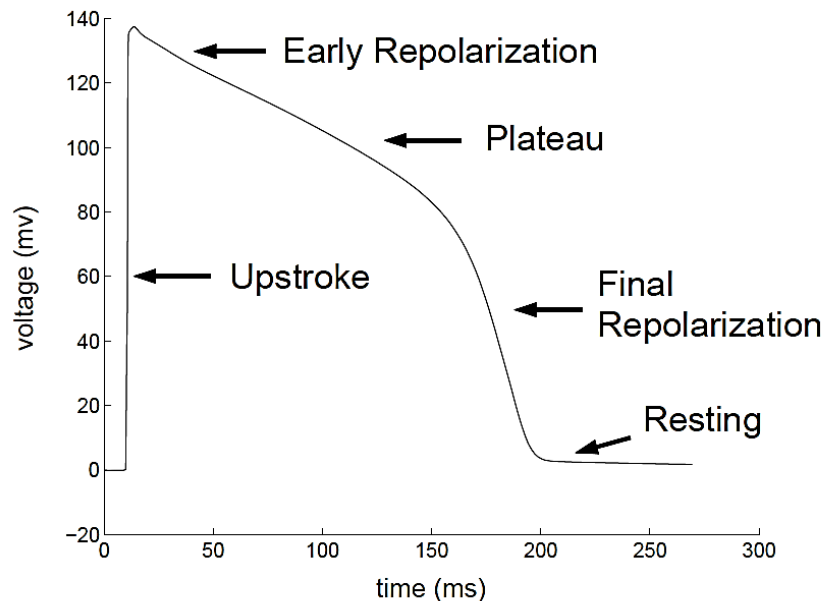


Fig. 1. Major AP phases.

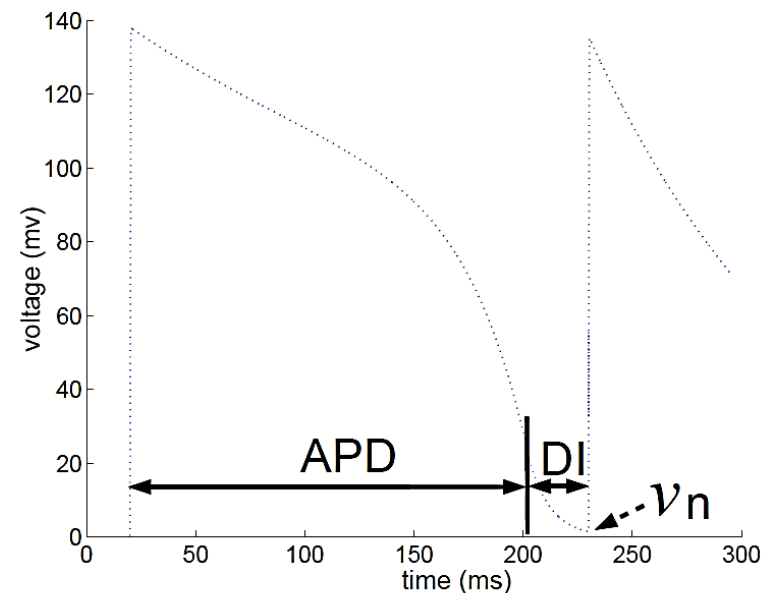


Fig. 2. APD and DI time periods.

P Ye, E Entcheva, R Grosu, SA Smolka, “Efficient modeling of excitable cells using hybrid automata”, International Conference Computational Methods in Systems Biology, 2005

Hodgkin-Huxley Model

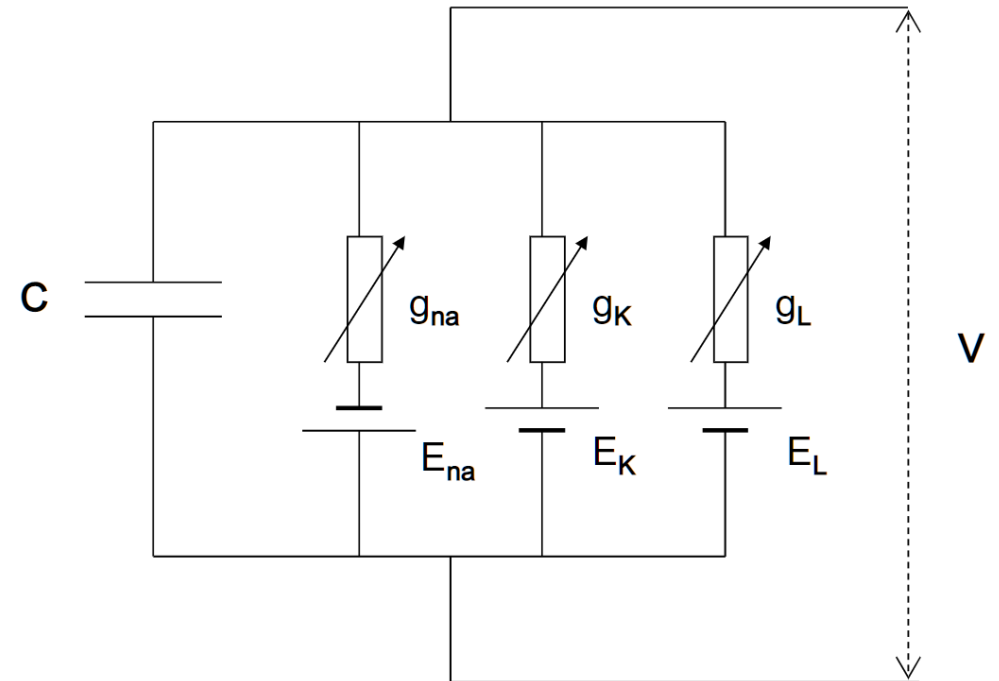
$$C\dot{V} = -\bar{g}_{Na}m^3h(V - E_{Na}) - \bar{g}_Kn^4(V - E_K) - \bar{g}_L(V - E_L) + I_{st}$$

- C : Cell capacitance
- V : Trans-membrane voltage
- g_{Na} , g_K , g_L : Maximum channel conductance
- E_{Na} , E_K , E_L : Reversal potential
- m , n , h : Ion channel gate variables
- I_{st} : Stimulation current

Mathematical model

Circuit model

Circuit for Hodgkin-Huxley Model



Modeling complex biological systems

- Technique:
 - **Hybrid Automaton (HA)** includes both discrete and continuous (differential equations) dynamical variables
 - A finite state machine
 - Digital and analog components can be modeled and analyzed
- Simple example for use of an HA application: room-thermostat-heater system
 - state of the heater (on/off) - discrete
 - Temperature evolves - continuous

(https://en.wikipedia.org/wiki/Hybrid_automaton)

Introduction to the basics of electronics

Principles and basic elements

Ohm's law

Kirchhoff's circuit laws

Applications

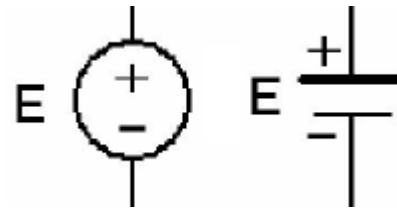
Principles and basic elements

- Conductor
 - allows the movement of electrons
- Insulator
 - does not allow the movement of electrons
- Electromotive Force - EMF
 - the force that moves the electrons through the conductor
 - unit of measure: Volt (V) – *research Alessandro Volta and its findings*
- Current
 - Flow of electrons
 - unit of measure: Ampere (A) - *research André-Marie Ampère and its findings*

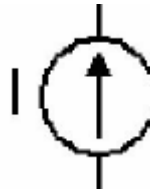
- Voltage source
 - has two terminals (+ and -)
 - Constant voltage or direct current (e.g., battery)
 - Alternating voltage or current (e.g., socket)

- Symbols

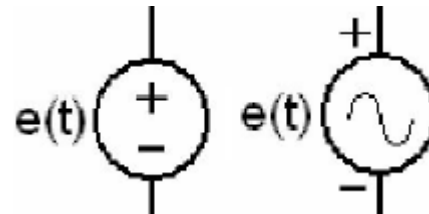
- DC voltage source



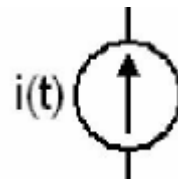
- direct current source



- variable voltage source



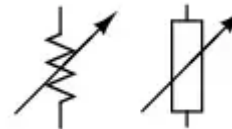
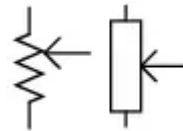
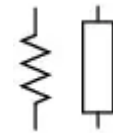
- variable current source



- Resistor
 - restrict the flow of electric current
 - unit of measure: Ohm (Ω)
 - electrical energy is converted to heat when current flows through a resistor
 - classification according to the variation of the electrical resistance
 - Fixed - has only one value and never changes (except through temperature, age, etc.)
 - Variable resistors - the value of the electrical resistance can be changed by the user
 - Variable resistors (potentiometers): a wide range of values, but is normally only able to handle electricity to a limited extent
 - Rheostat: a limited range of values and a high current carrying capacity

- Symbols:

- Fixed
- Potentiometer
- Rheostat

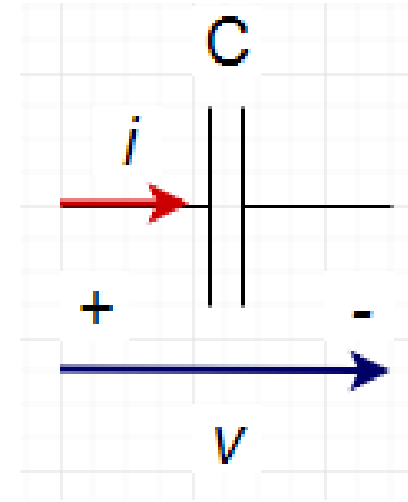


- Capacitor

- stores electric charge; measure of how much charge they can hold
- unit of measure: Farad (F)

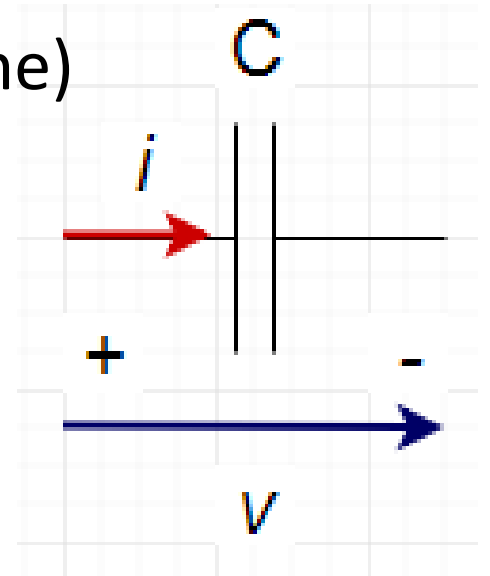
Charge, $Q = C \times V$

Energy, $E = \frac{1}{2}QV = \frac{1}{2}CV^2$

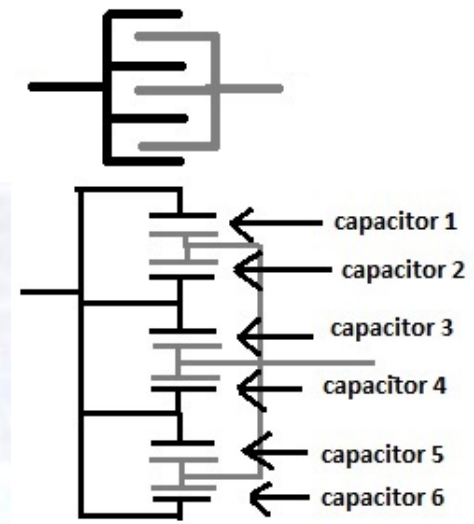


- Q – the amount of electrical charge stored on the two armatures/plates - coulombs (C)
- C – the electrical capacity of the capacitor indicates how much electrical charge it can store - farads (F)
- V – the value of the voltage applied between the capacitor terminals - volts (V)
- E - capacitors are also storing energy - joules (J)

- $i = \frac{dQ}{dt}$
- i – the current through the capacitor; *instantaneous* current (the amount of current at a specific point in time)
- Capacitor operation equation
 - $i = C \times \frac{dv}{dt}$
- $\frac{dv}{dt}$ instantaneous rate of voltage change (volts per second)



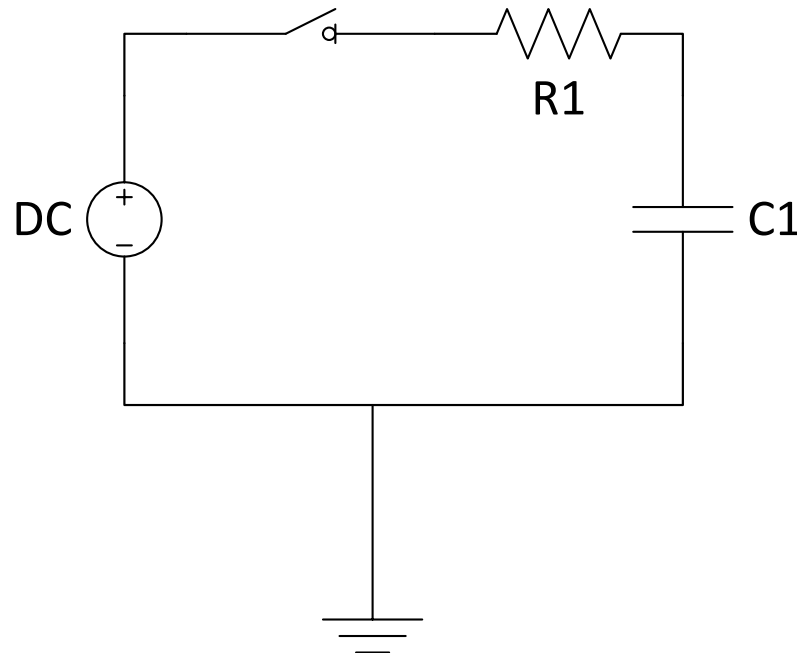




Interleaved capacitor
Condensator intercalat

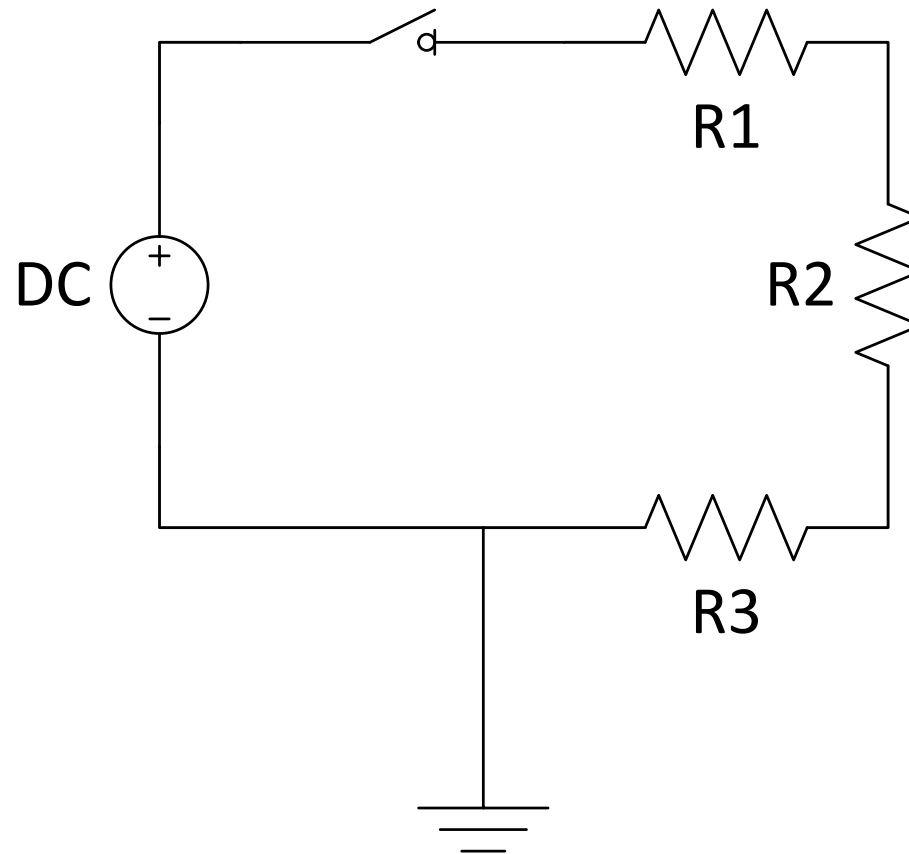
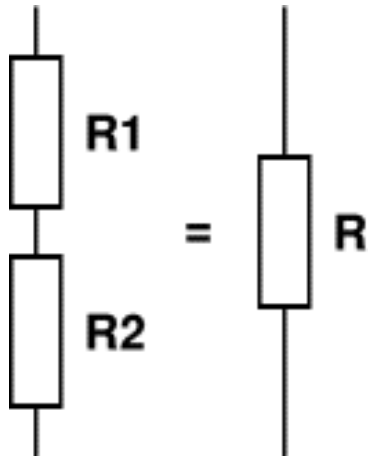
Principles and basic elements

- Electrical circuit
 - path for electron flow
 - open: the path is interrupted; current does not flow
 - closed: the path is not interrupted; the current flows



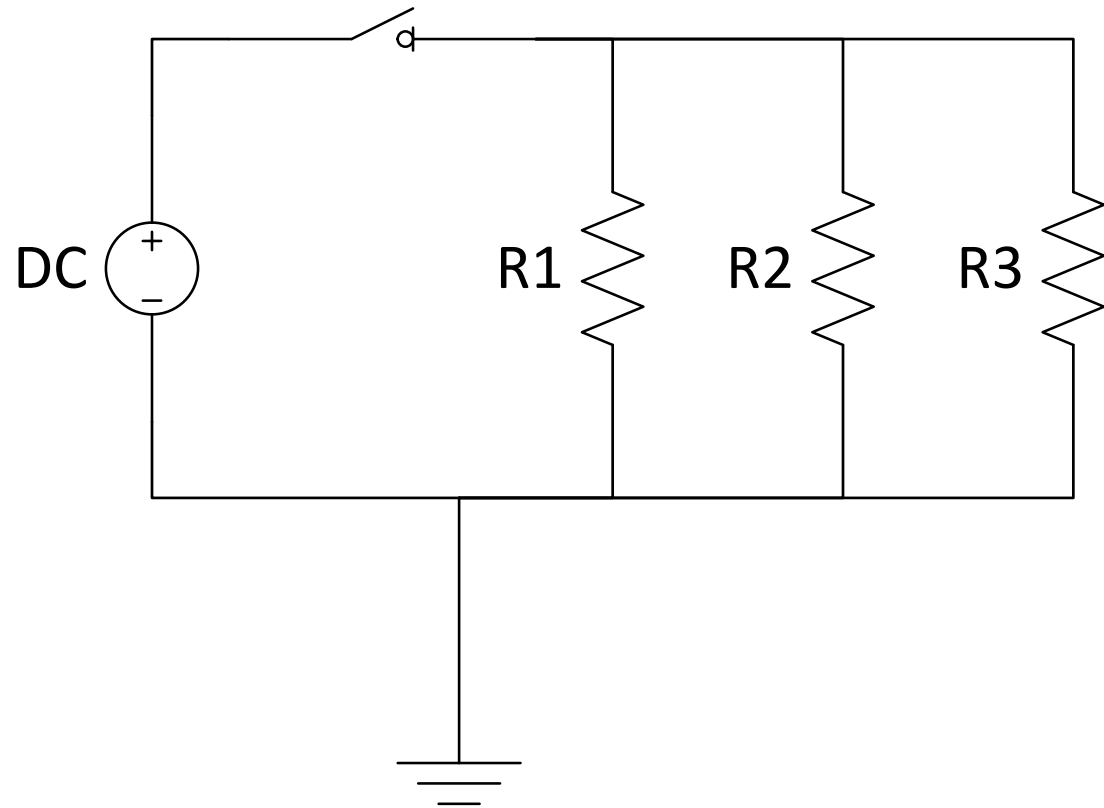
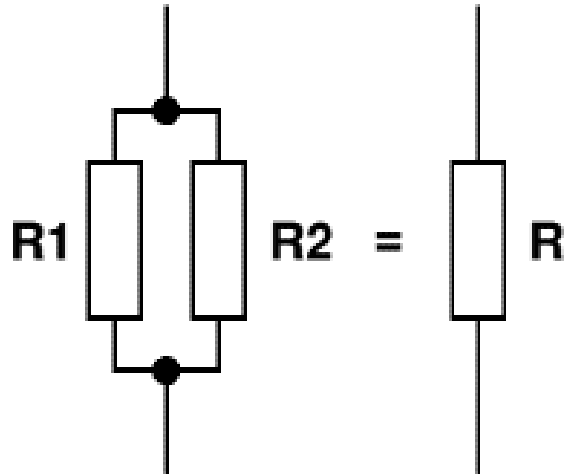
Resistors connected in Series

- $R_e = R_1 + R_2 + \dots + R_n$



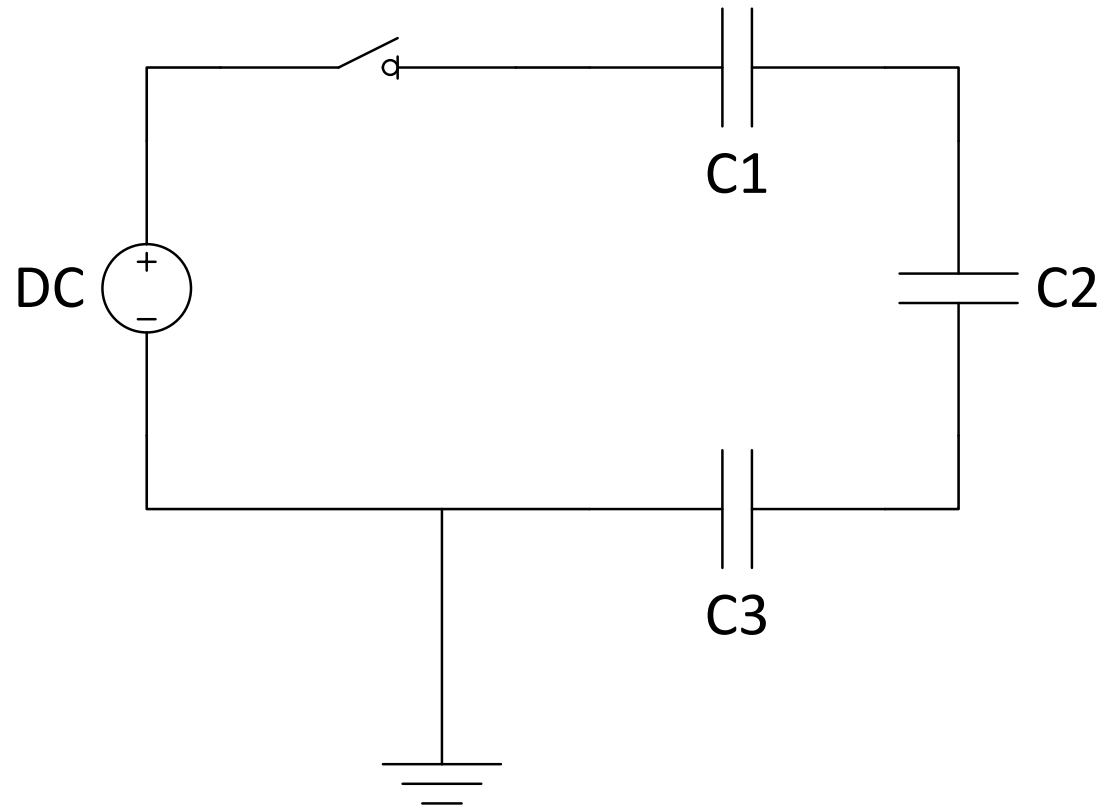
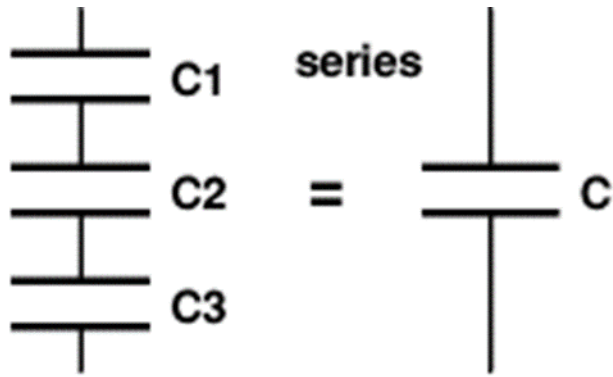
Resistors connected in Parallel

- $1/R_e = 1/R_1 + 1/R_2 + \dots + 1/R_n$



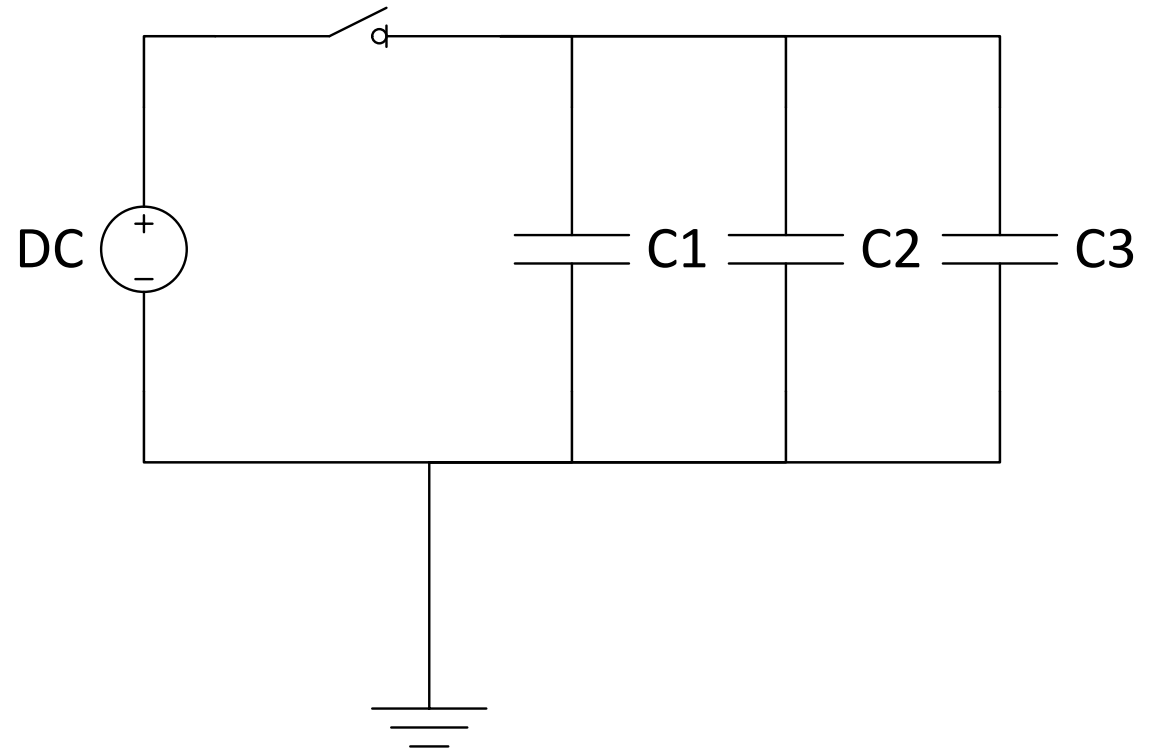
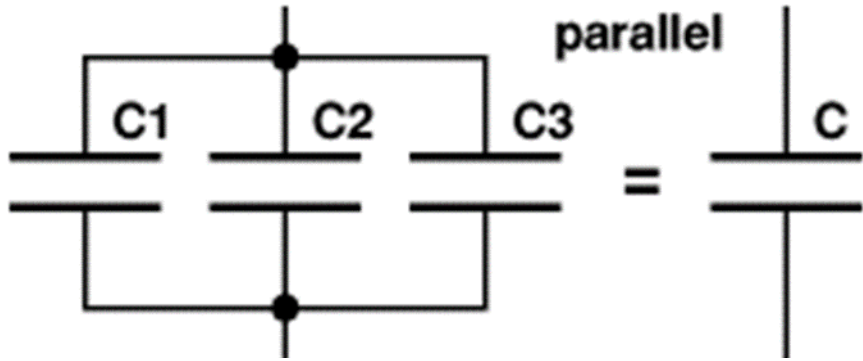
Capacitors in Series

- $1/C_e = 1/C_1 + 1/C_2 + \dots + 1/C_n$

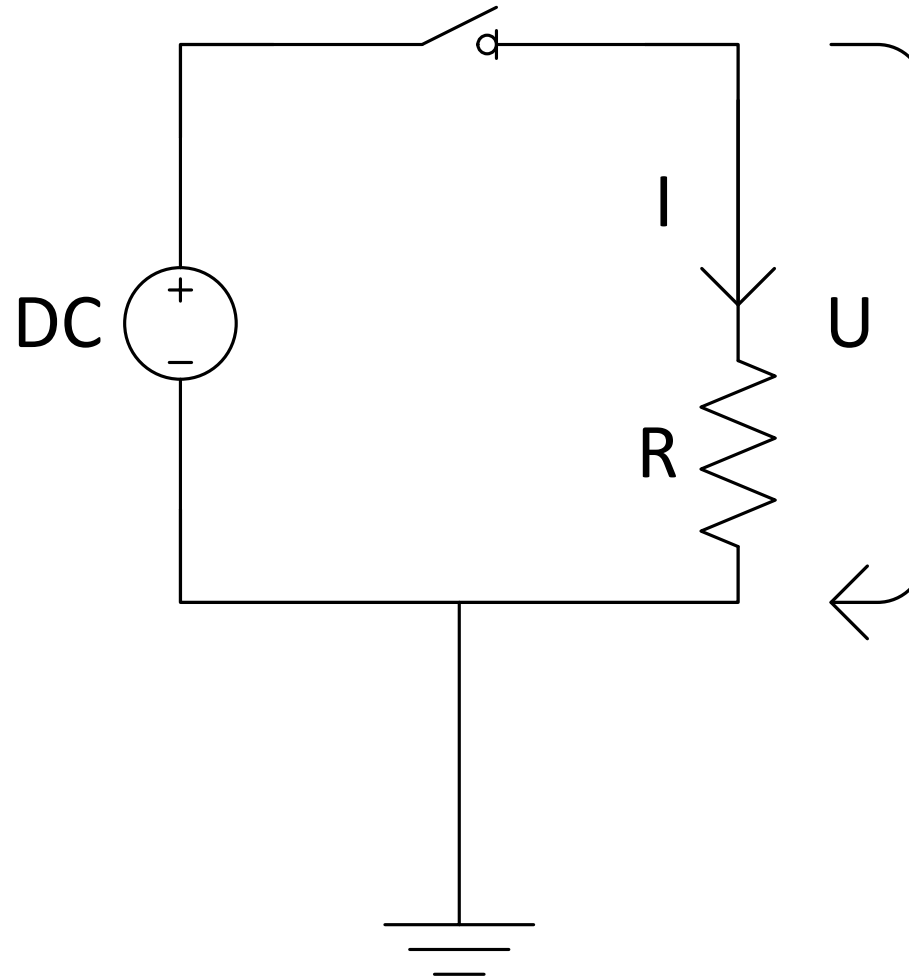
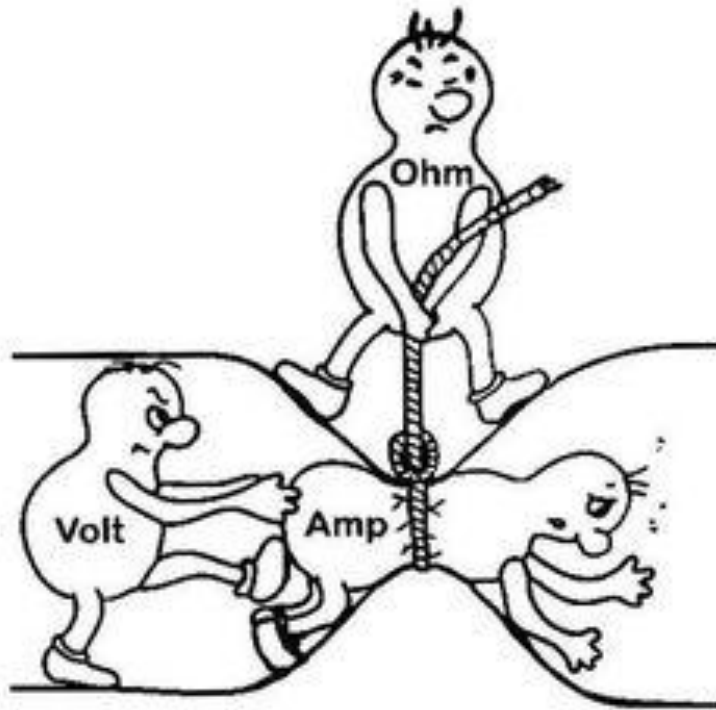


Capacitors in Parallel

- $C_e = C_1 + C_2 + \dots + C_n$



Ohm's law



- $U = I \cdot R$
- $I = U / R$
- $R = U / I$

Kirchhoff's circuit laws

- Current Law

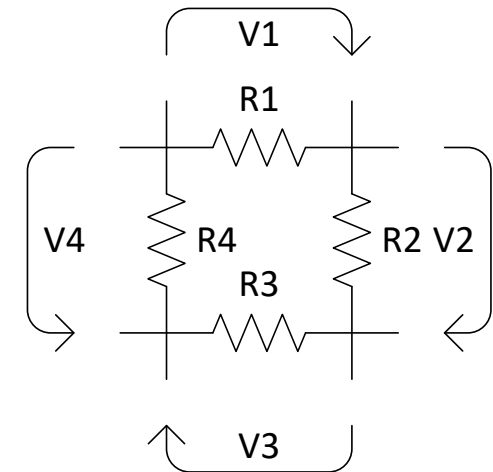
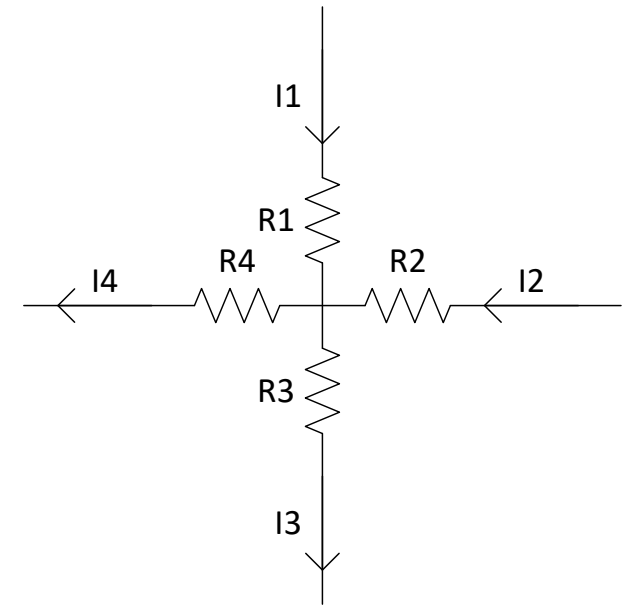
- The total current or charge entering a junction or node is exactly equal to the charge leaving the node as it has no other place to go except to leave, as no charge is lost within the node

$$\sum_{k=1}^n I_k = 0$$

- Voltage Law

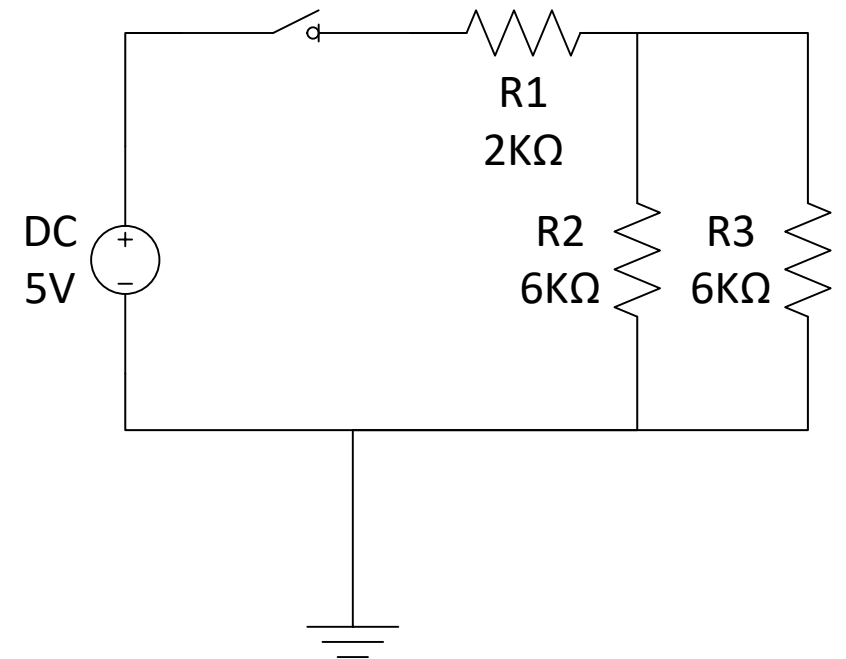
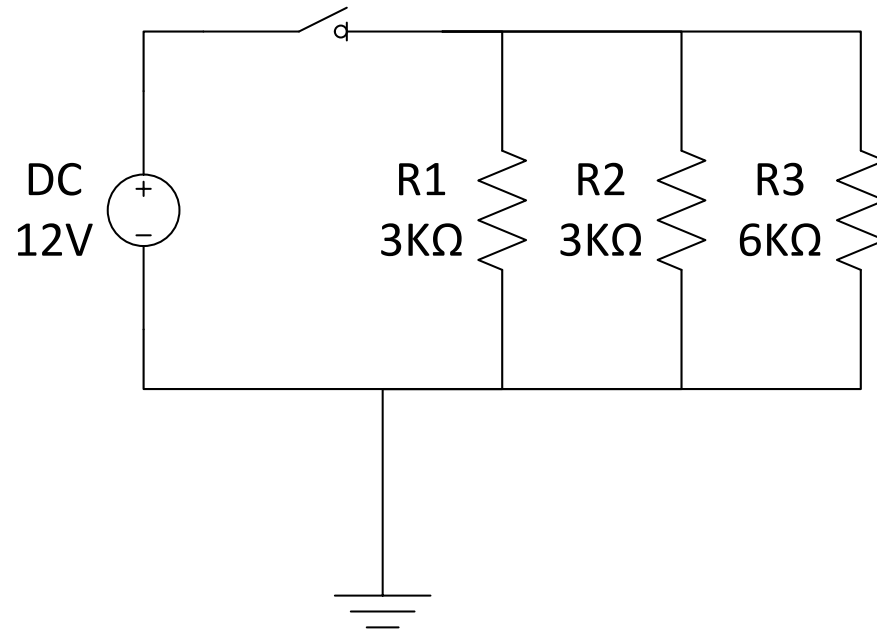
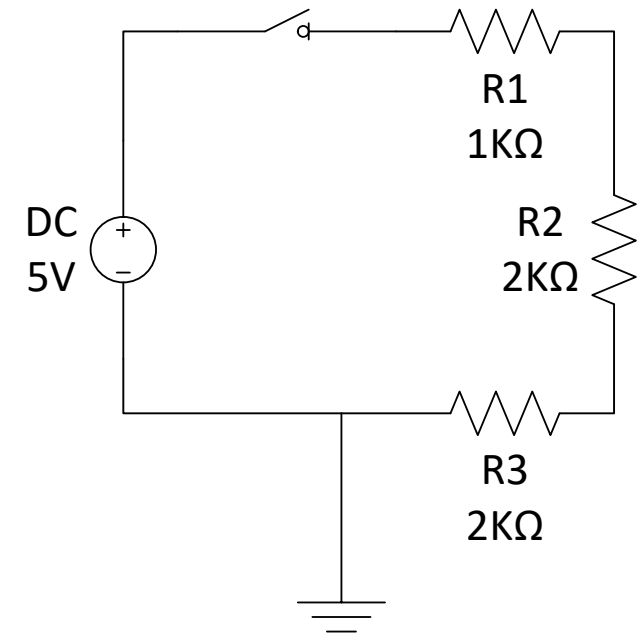
- in any closed loop network, the total voltage around the loop is equal to the sum of all the voltage drops within the same loop" which is also equal to zero

$$\sum_{k=1}^n V_k = 0$$



Applications

- For the circuits below calculate the voltage drops and currents through the resistors



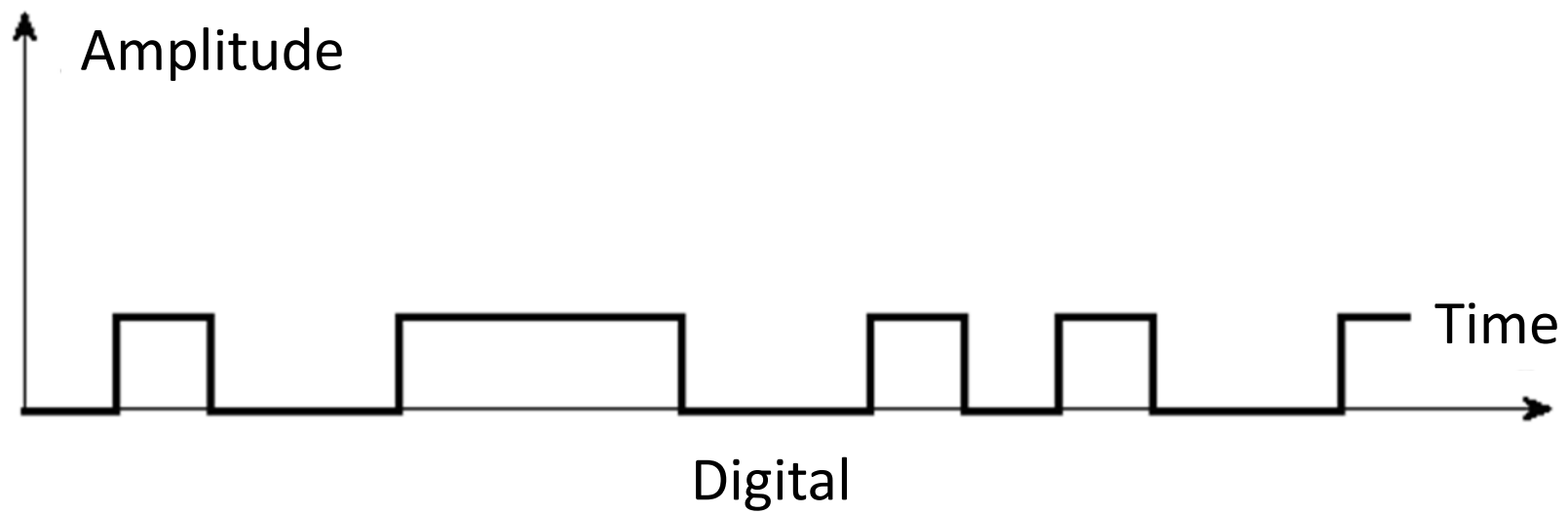
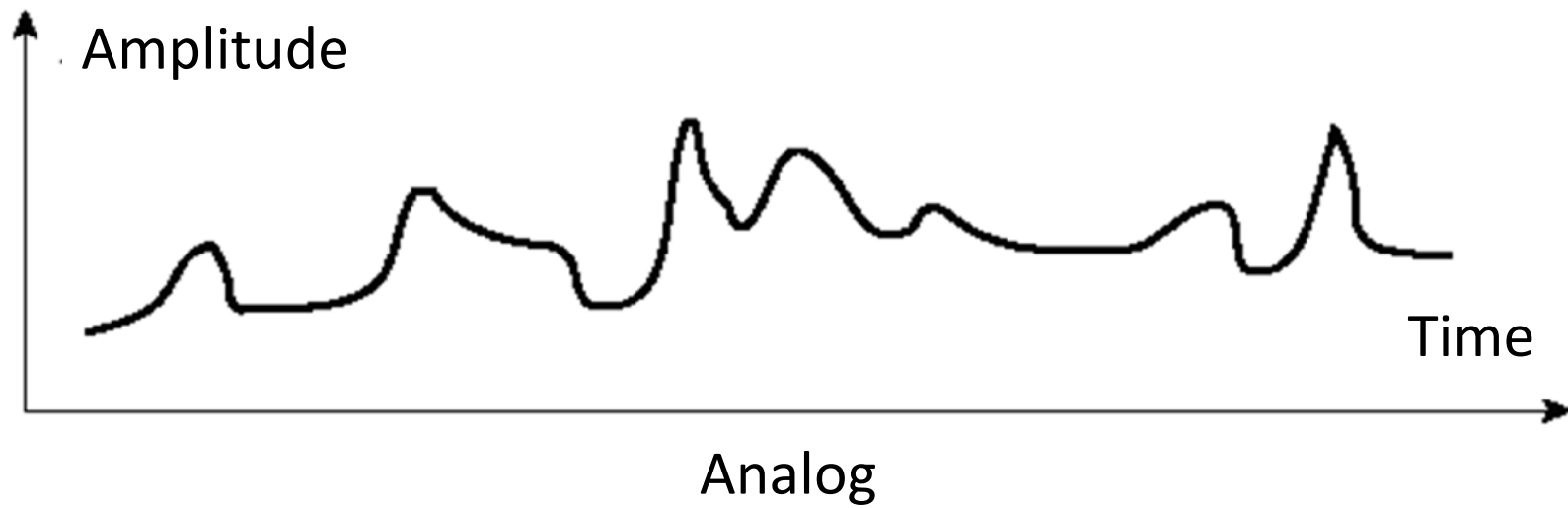
Chapter 1: Basic presentation for: electric signals, pulse signal, RC linear circuits, pulse transformers

Content

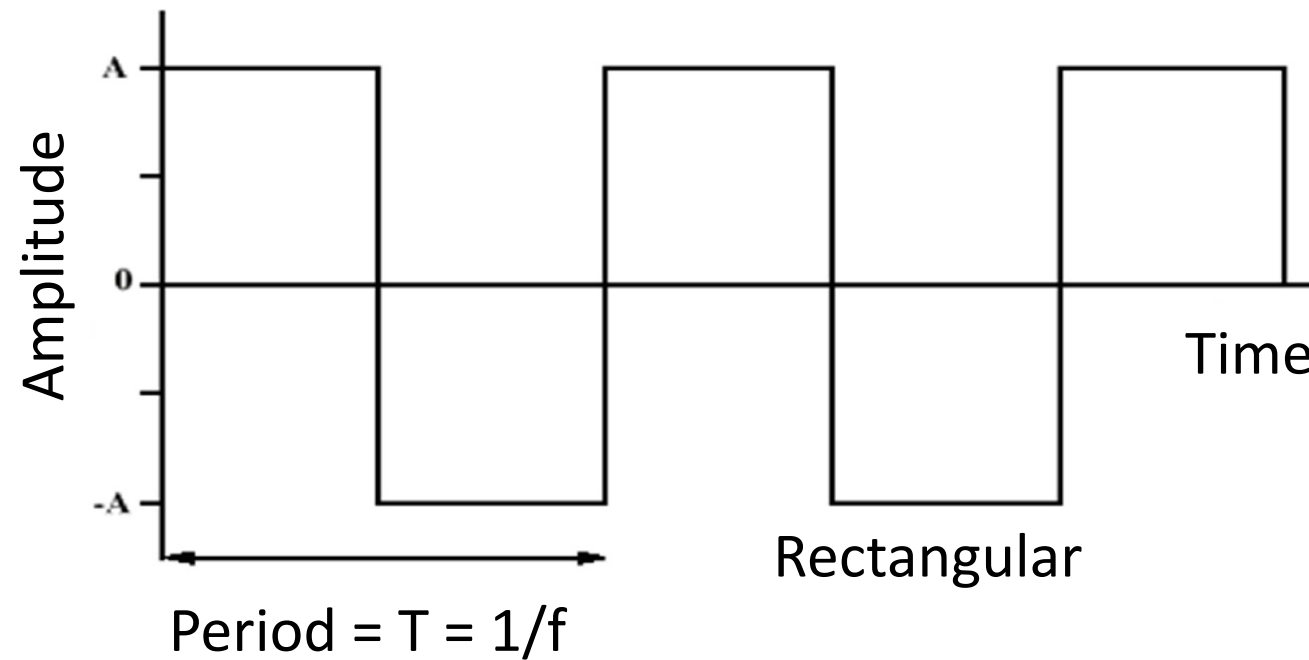
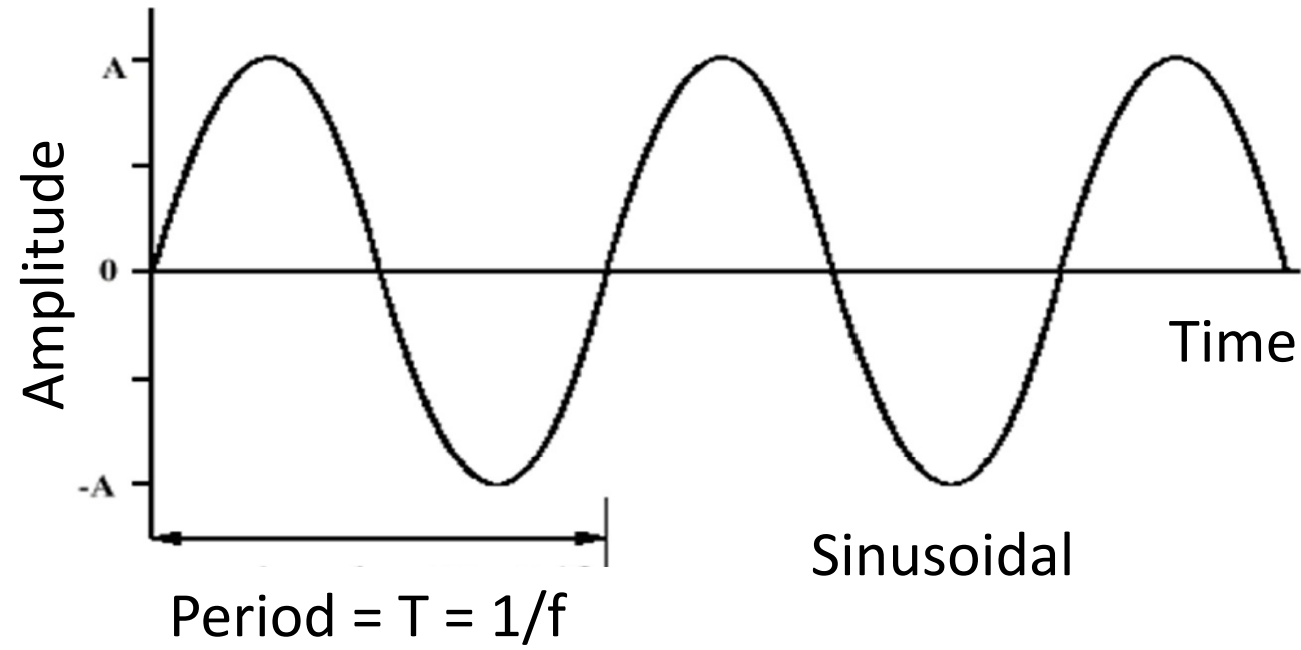
- Electric signal - basics
- Electric pulse: definition, parameters, pulse generating using elementary signals
- RC circuits (pulse transformers)
- RC circuits response to elementary signals

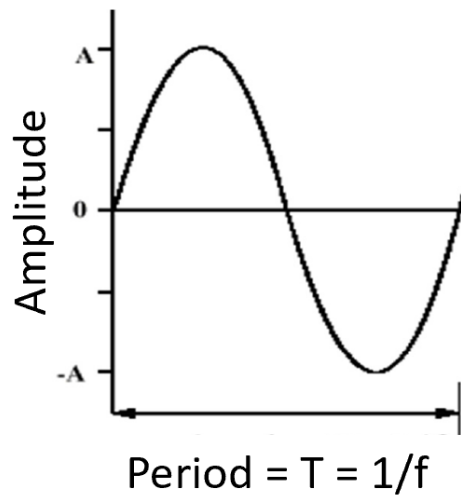
Electric Signals - basics

- Defined as function of one/more variables, bearing information about the physical nature of a phenomenon
- Classification after number of variables:
 - unidimensional
 - multidimension
- Classification after behaviour in time domain:
 - continuous – continuous variation; may present any value
 - discrete – waveform maintains a constant level, then switches suddenly to another level



- Classification after symmetry:
 - *even* – symmetrical with regard to the vertical axis or to the time origin
 - *odd* - anti-symmetrical with regard to time origin
- Classification after periodicity:
 - *periodical*: $v(t) = v(t \pm n T_0)$ for any value of t .
 T_0 means signal's period.
 Reverse of period defines signal's frequency: $f = 1/T_0$.
 Based on frequency, there is defined angular frequency: $\omega = 2\pi f$
 - *non-periodical*: $v(t) \neq v(t \pm nT_0)$





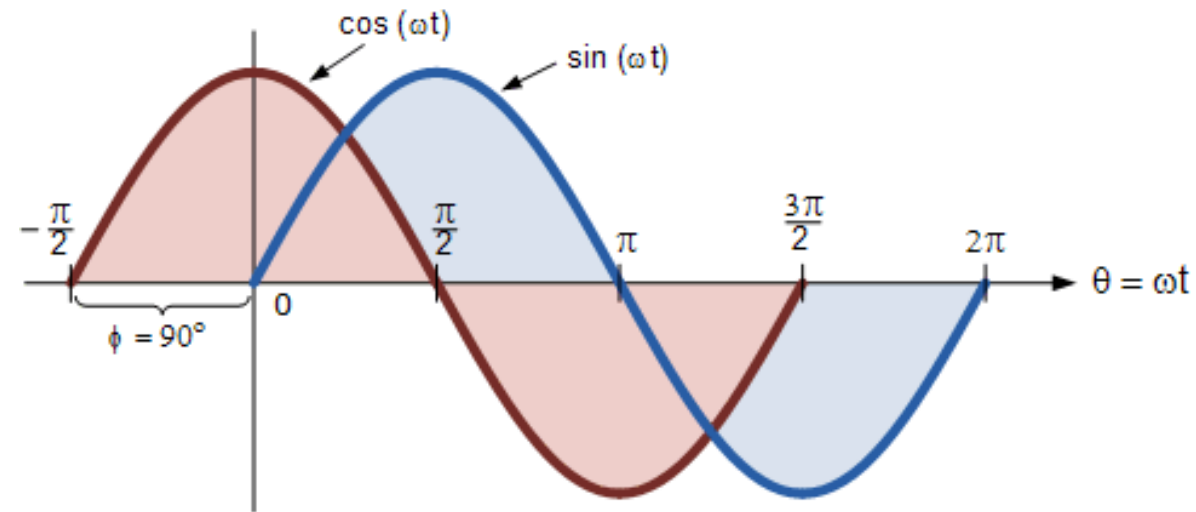
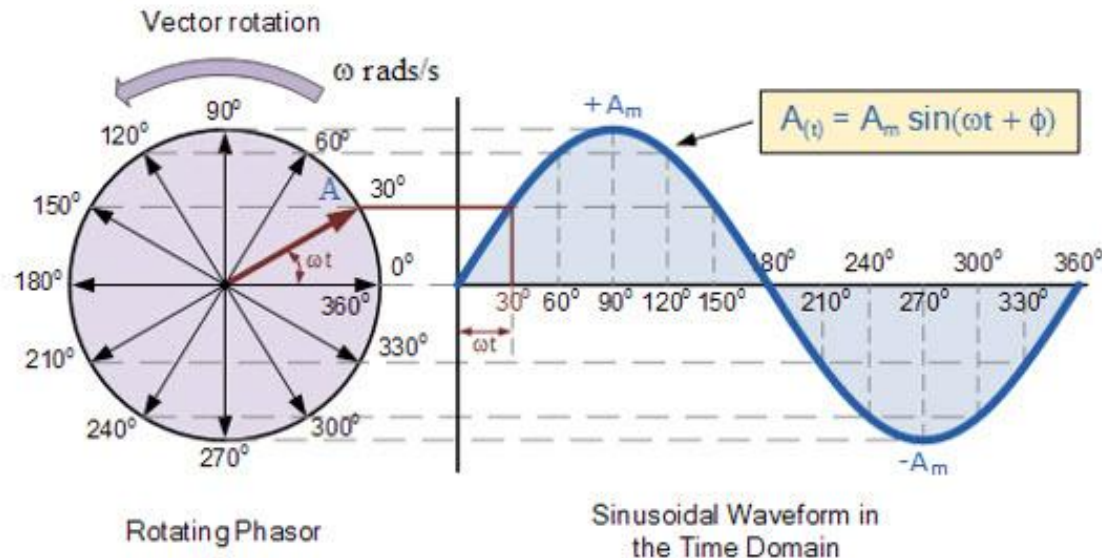
Elementary signals

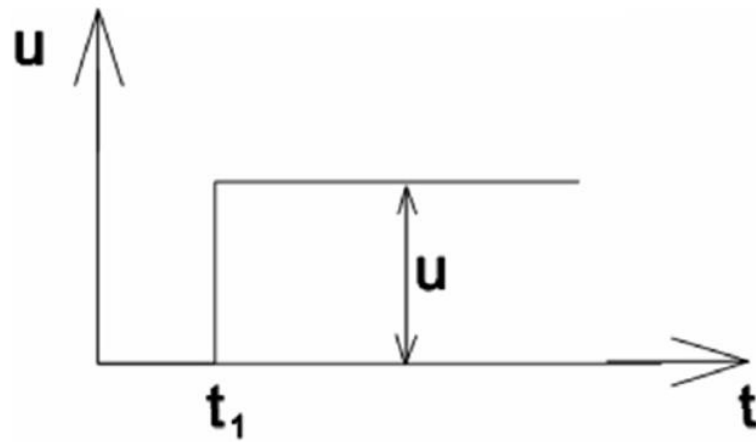
- Sine / Cosine**

$$v(t) = A \cos(\omega_0 t + \phi) = A \cos(2\pi f_0 t + \phi)$$

A – amplitude, ω_0 – angular frequency, f_0 – frequency, ϕ - phase

Phasor Diagram of a Sinusoidal Waveform

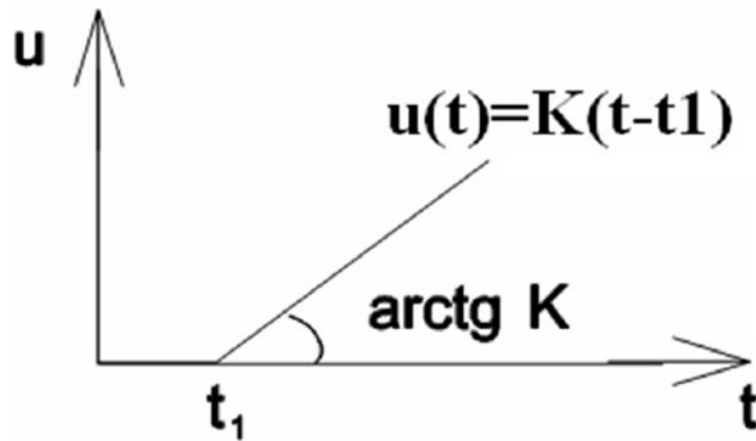




- **Step signal**

$$u(t) = U, \text{ for } t \geq t_1$$

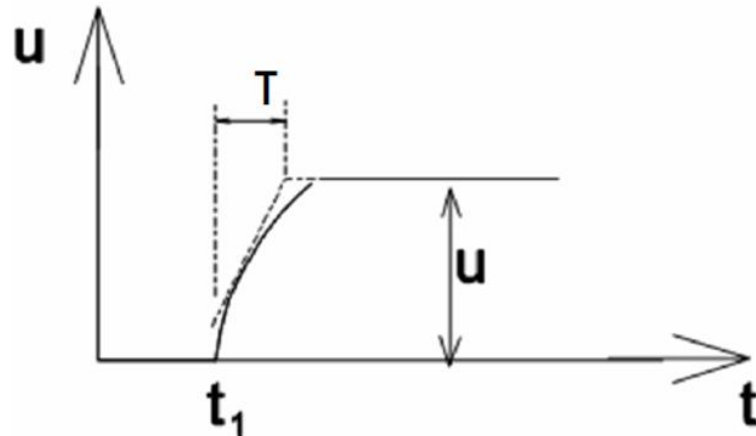
$$u(t) = 0, \text{ for } t < t_1$$



- **Ramp**

$$u(t) = k \cdot (t - t_1), \text{ for } t \geq t_1$$

$$u(t) = 0, \text{ for } t < t_1$$



- **Exponential**

$$u(t) = U(1 - e^{-(t-t_1)/\tau}), \text{ for } t \geq t_1$$

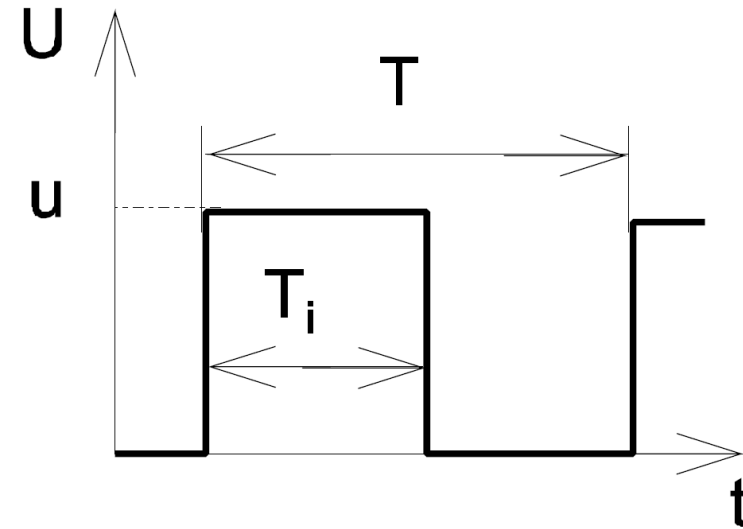
$$u(t) = 0, \text{ for } t < t_1$$

τ – time constant of signal

Definition of pulse

- PULSE: a voltage or current signal that has its value different from a constant value (that can also be zero), only for a temporary short enough period, smaller or comparable with the transitory period of the circuit that is transmitting the signal

Shape of an ideal pulse



U – pulse amplitude

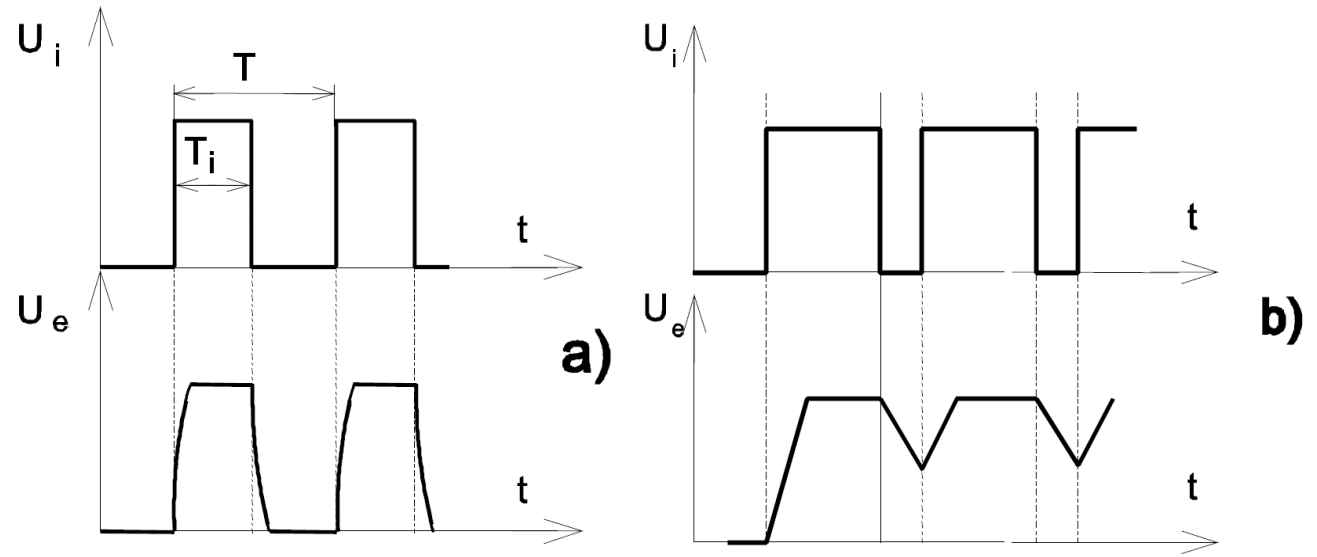
T_i – pulse duration (width)

T – pulse repetition period

Relation between the transitory regime of a circuit and the pulse signal

- Any state switch of a circuit implies generation of a transitory regime, taking a bounded time τ
- For a pulse signal, the switch times between levels must be lower or comparable with the period of the transitory regime
- For a pulse the repetition period T must be much greater than the duration of the transitory regime

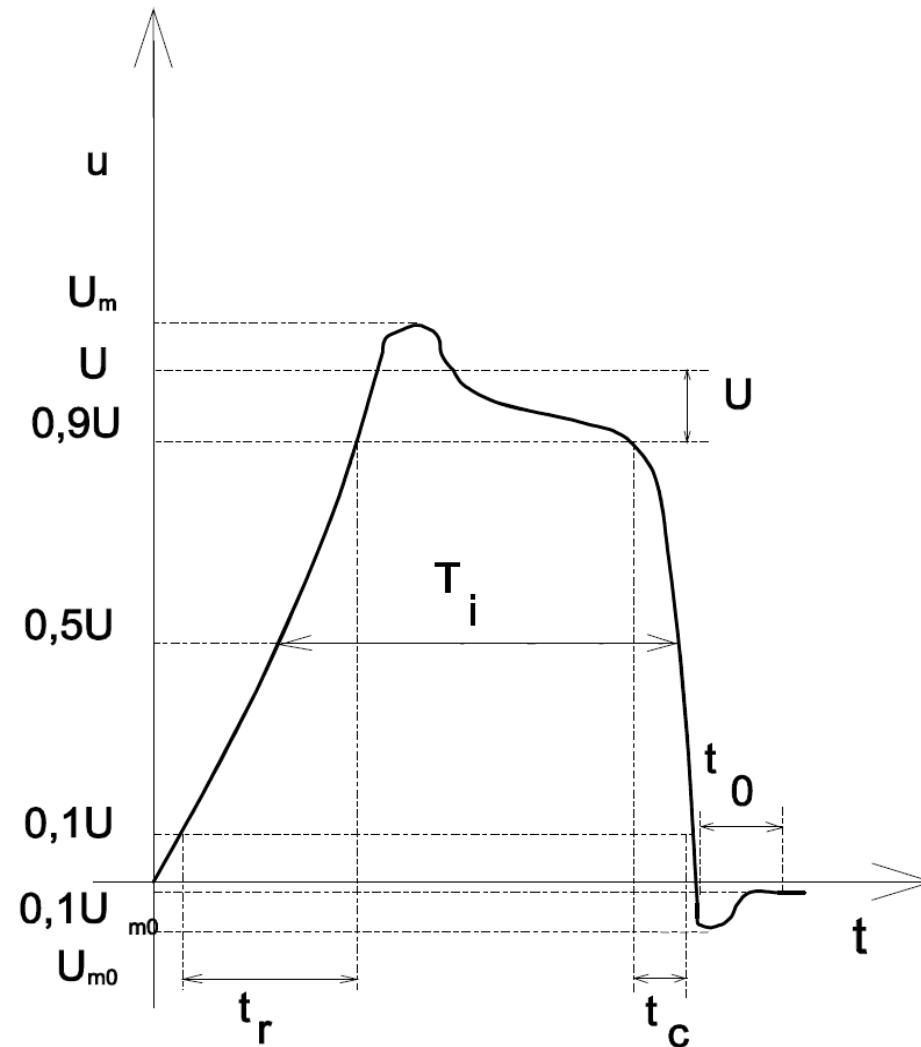
Pulse period versus transitory time



Right values

Wrong definition

Shape of a real pulse



U – pulse amplitude

U_m – over-exceeding amplitude (overshoot)

U_{m0} – under-exceeding amplitude (preshoot)

ΔU – voltage drop on a level (ringing)

T_i – pulse width

t_r – rising time - time of leading (anterior) edge)

t_c – fall-down time - time of trailing (posterior) edge

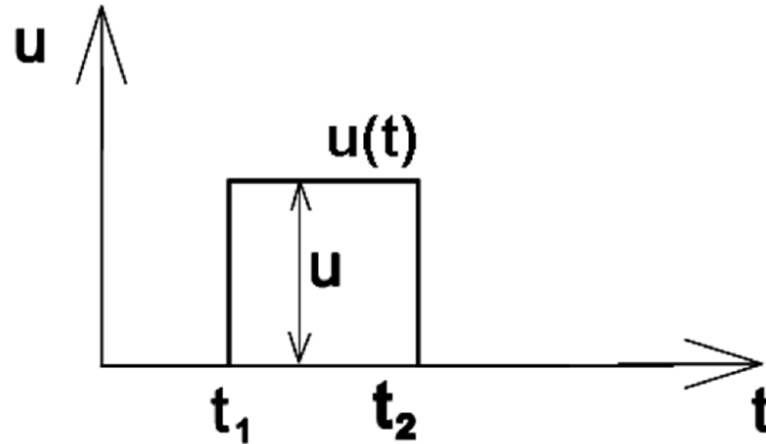
t_0 – reverse return time

T – repetition period

f_u – space factor: $f_u = T_i/T$

f – repetition frequency: $f = 1/T$

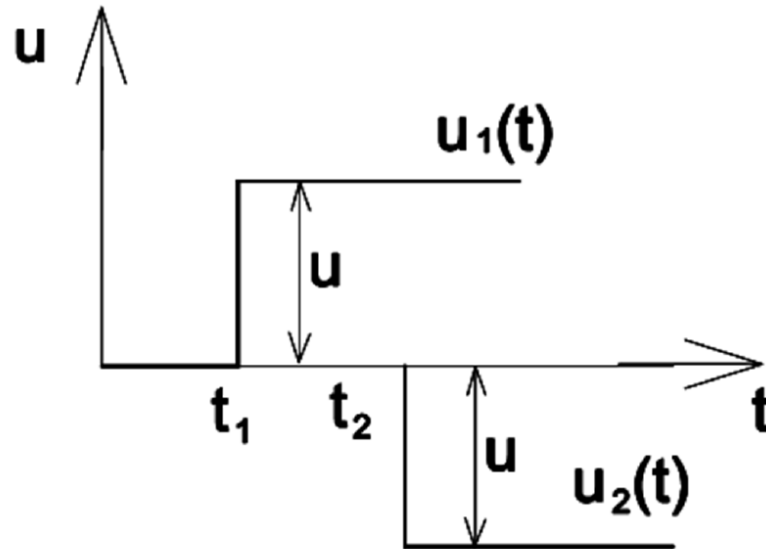
Generating a pulse by composing some elementary signals



$$u(t) = u_1(t) + u_2(t)$$

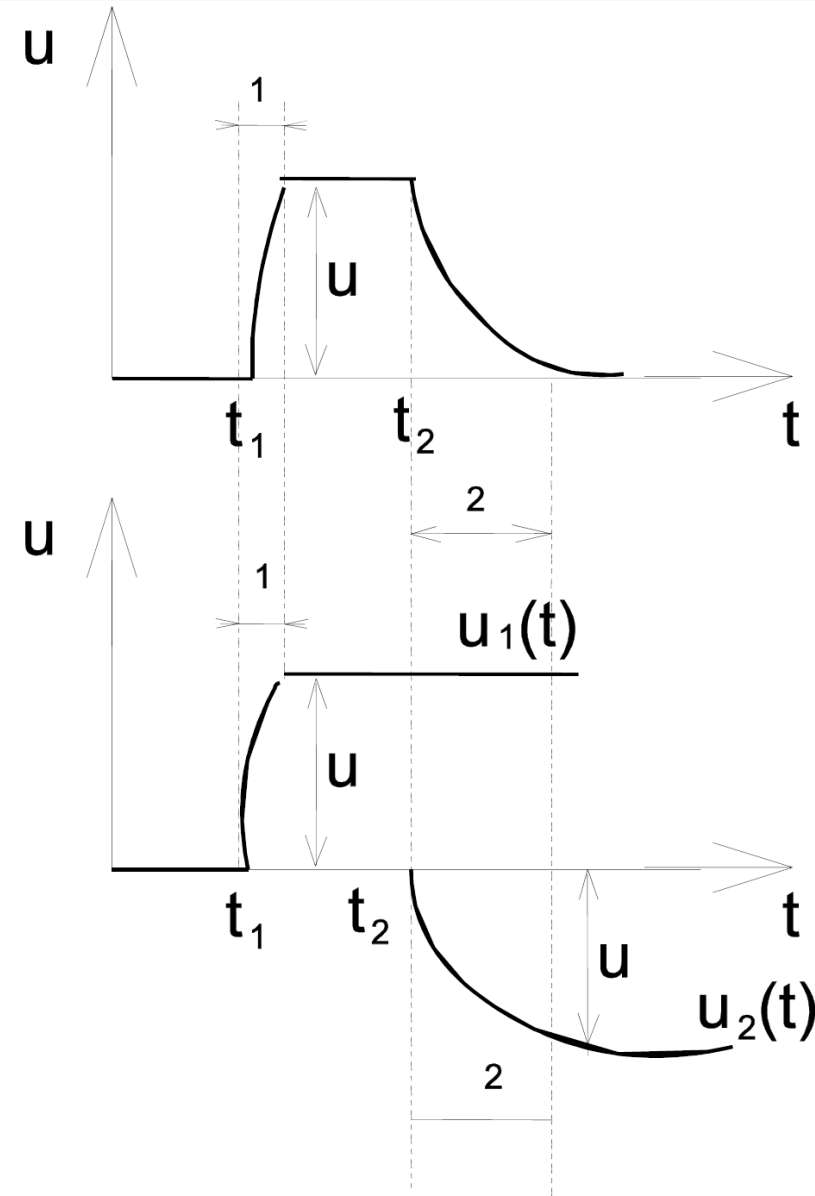
$u_1(t)$ – positive single step signal

$u_2(t)$ – negative single step applied at t_2



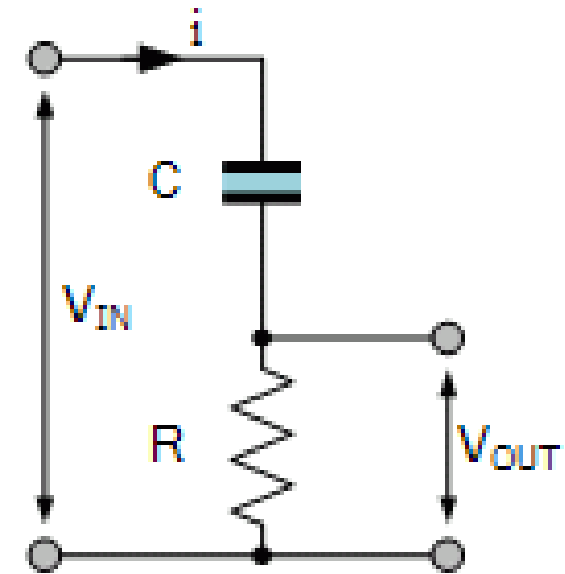
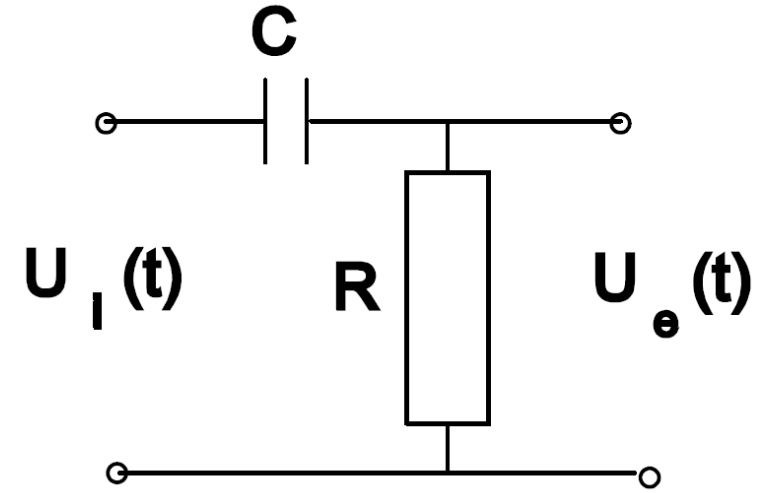
Generating a pulse by composing elementary signals

- Example with two exponential signals with different signal time constants



RC high-pass filter

- **capacitive reactance** * varies inversely proportional with the frequency
- circuit behaves like a voltage divider whose dividing ratio depends on frequency
- resistance times the capacitance ($R \times C$) is the **time constant**;
 - it is inversely proportional to the **cut-off frequency**

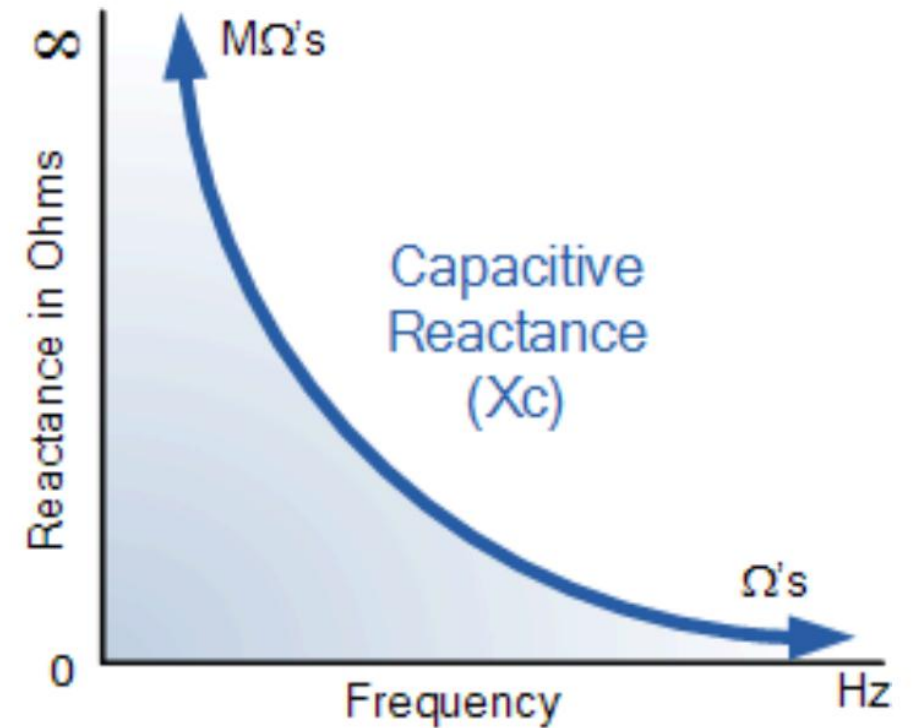


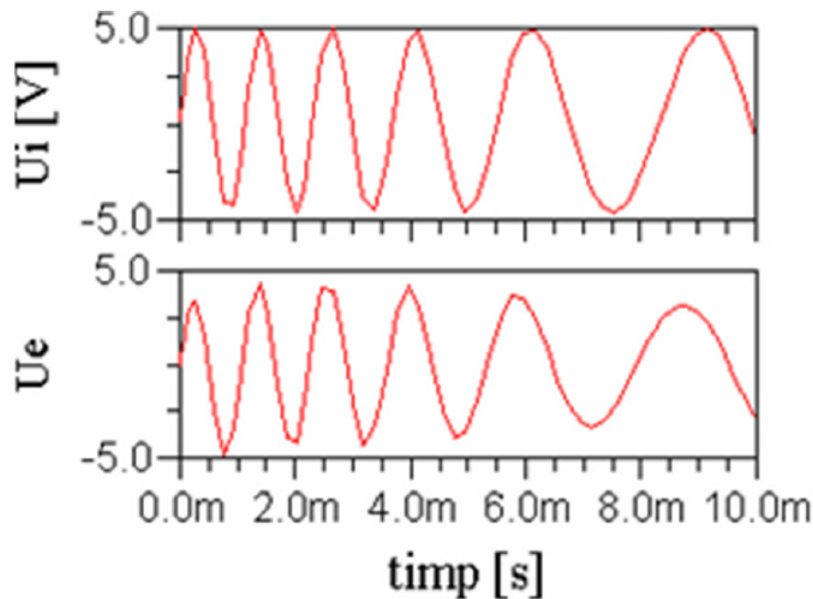
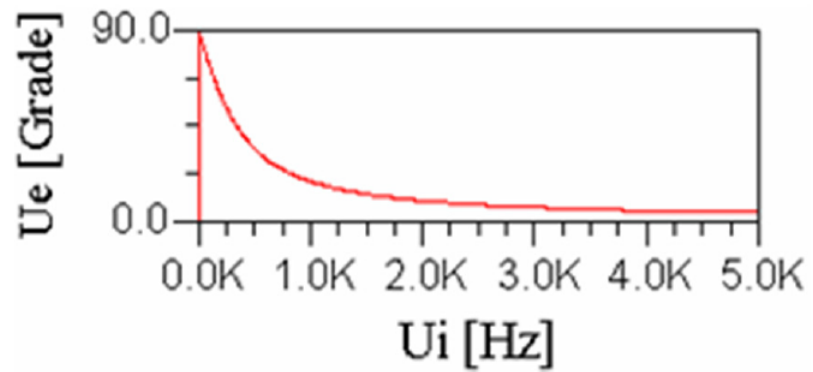
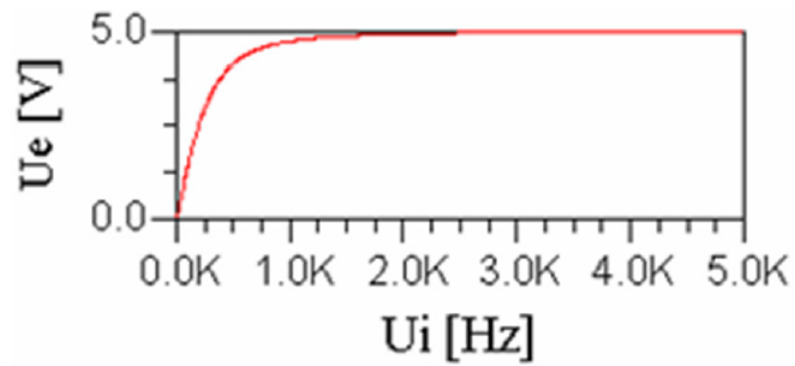
*capacitive reactance

- X_c = reactance in ohms
- f = frequency in hertz
- C = capacitance in farads

$$X_c = \frac{1}{2\pi fC}$$

- large at low frequencies and small at high frequencies
 - for DC (zero frequency), X_c is infinite





Circuit Response for a sinus input signal

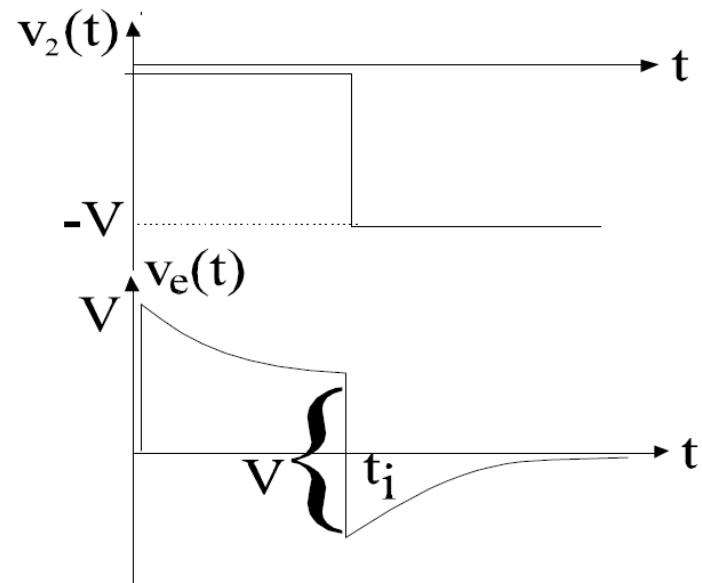
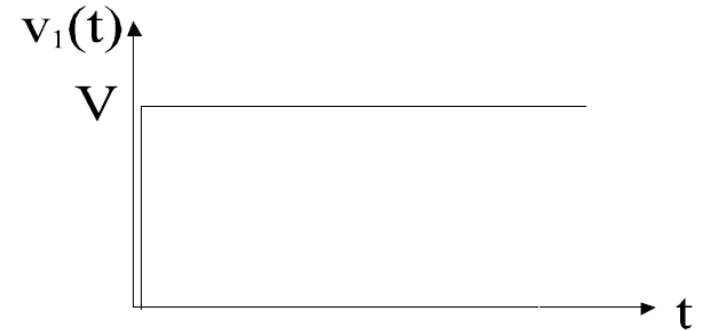
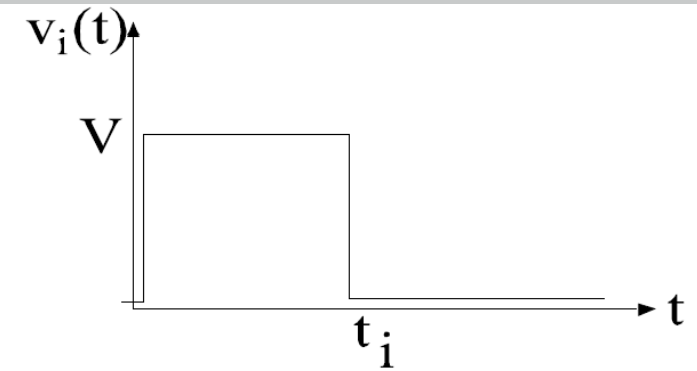
- $U_i = |U_i| e^{j\omega t}$, $\omega = 2\pi f$
- $U_e = |U_e| e^{j(\omega t - \phi)}$
- Circuit response is a sinus signal with attenuated amplitude $A(\omega)$ and a phase shift $\phi(\omega)$

$$A(\omega) = \frac{1}{\sqrt{1 + \left(\frac{1}{\omega RC}\right)^2}}$$

$$\phi(\omega) = \arctg\left(\frac{1}{\omega RC}\right)$$

RC high-pass filter: response at a pulse input signal

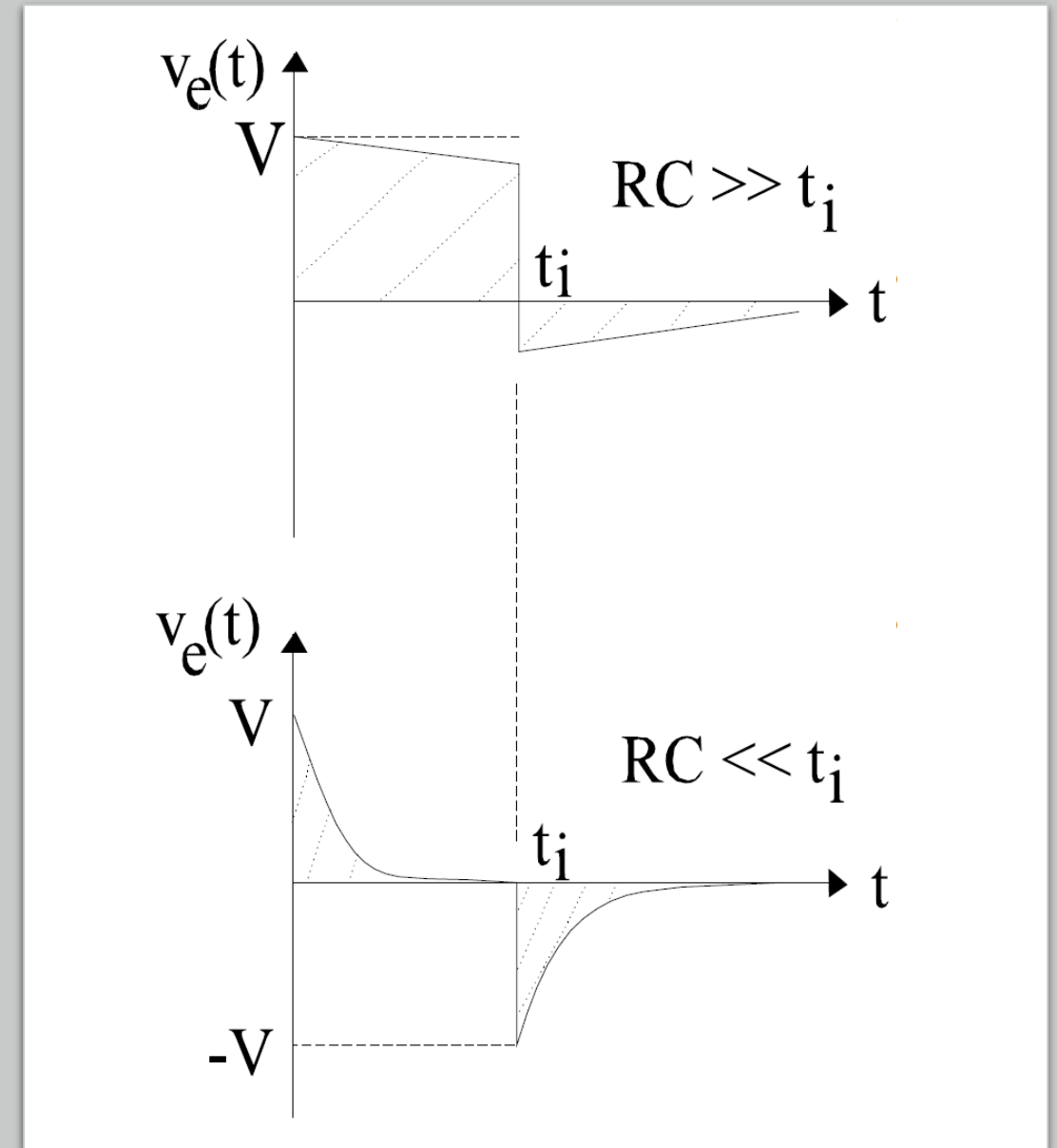
- Pulse generated using two step signals of amplitudes: $+V$ și $-V$, applied at moments $t=0$ and respectiv. $t=t_i$
- continuous (dc) component of the input signal does not appear at the output
- circuit is also called separation circuit, and is used for separating direct current circuits
- can produce trigger or spiked typed pulses for timing circuit applications



RC high-pass filter: response at a pulse input signal

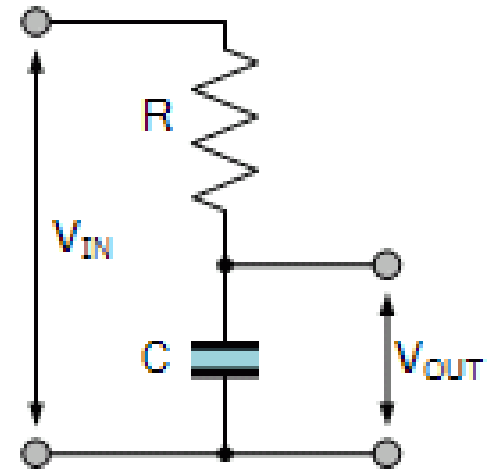
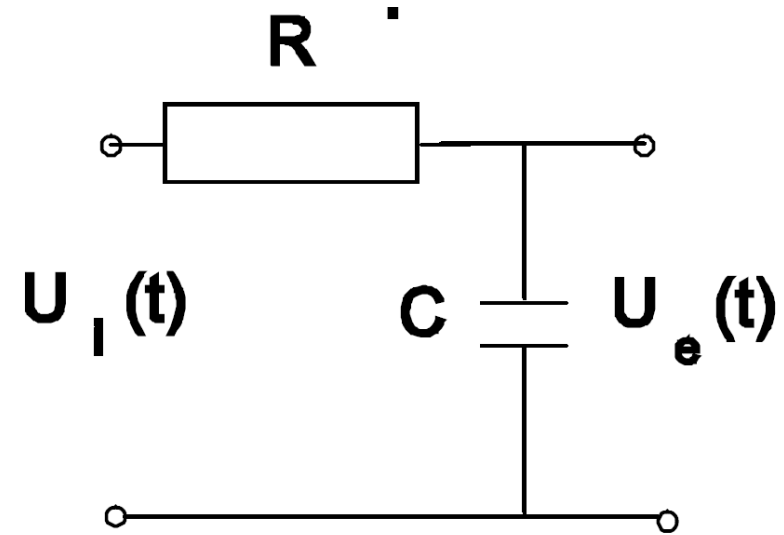
- the area above the abscissa is always equal to the area below the abscissa (time axis)
- to obtain negligible distortions, it is necessary to choose an RC time constant much higher than the impulse length t_i
- circuit can be used as a **differentiating circuit**, with the condition that the time constant is much smaller than the impulse length (*corresponds to the mathematical process of differentiation*)

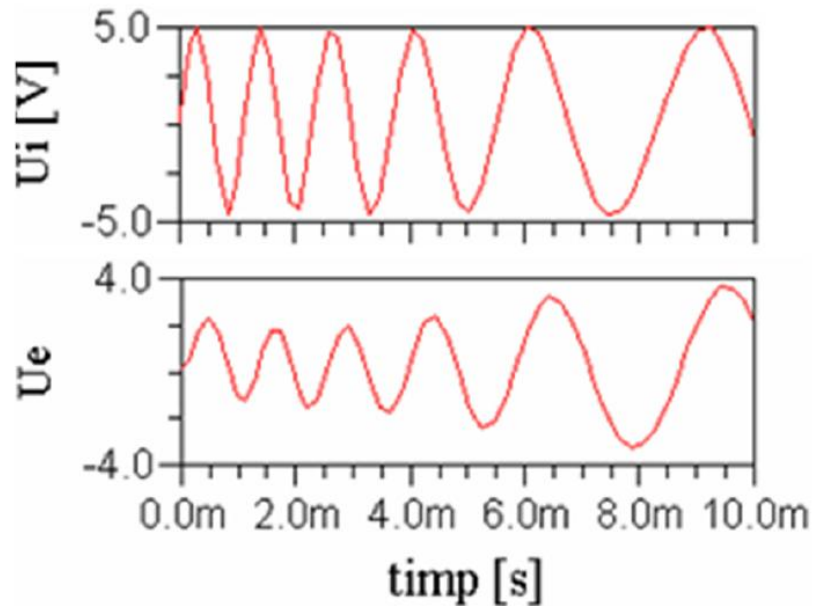
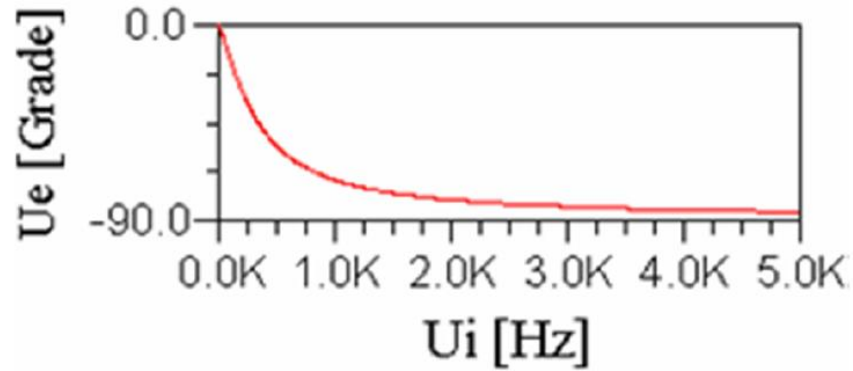
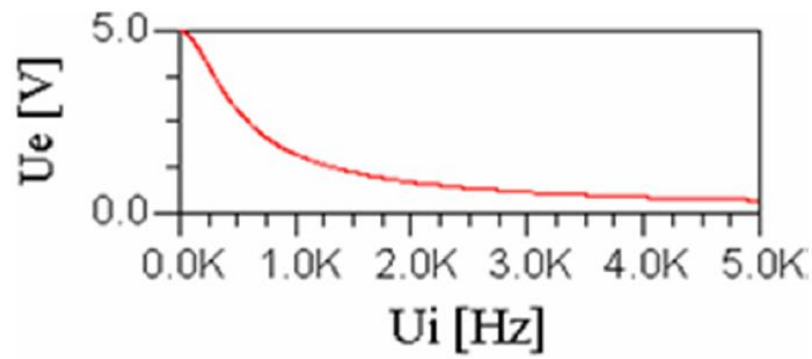
$$V_{OUT} = RC \frac{dV_{IN}}{dt}$$



RC low-pass filter

- Opposite to the high-pass filter
- harmonics with higher frequencies appear at the output with a higher attenuation than the low frequency harmonics





Circuit Response for a sinus input signal

- $U_i = |U_i| e^{j\omega t}$, $\omega = 2\pi f$
- $U_e = |U_e| e^{j(\omega t - \phi)}$
- Circuit response is a sinus signal with attenuated amplitude $A(\omega)$ and a phase shift $\phi(\omega)$

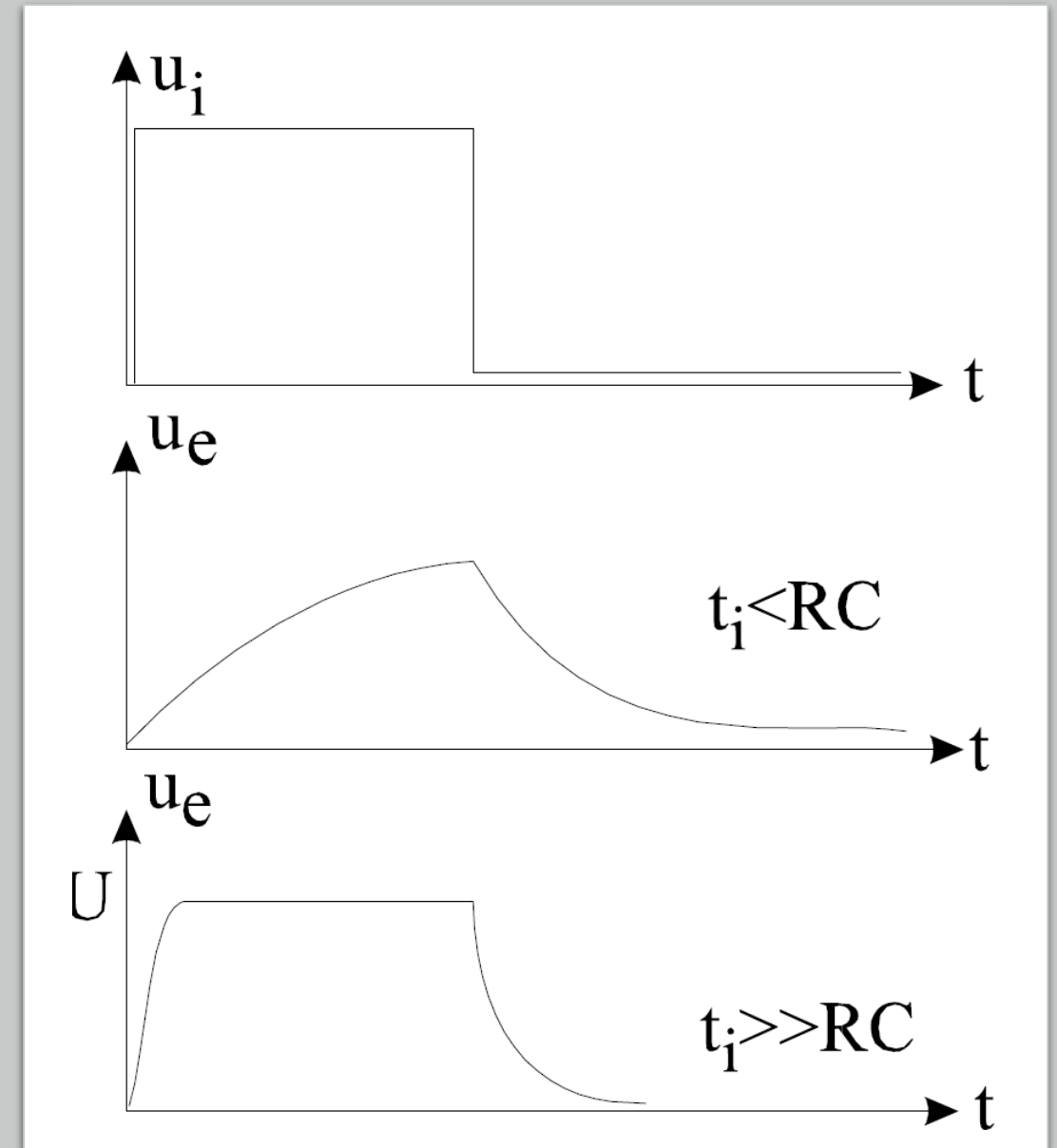
$$A(\omega) = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$\phi(\omega) = -\arctg(\omega RC)$$

Low-pass filter: Circuit response for a single step input signal

- In order for the distortions introduced by the circuit upon the impulse type input signal to be negligible, it is necessary for the circuit's elements to satisfy the relation $RC \ll t_i$
- If the voltage drop on the resistance much greater than on the capacitor: $U_C \ll U_R$, the response of the circuit will represent the integral of the input signal with respect to time
- circuit is used for pulse remaking, like an **integrator circuit** (*corresponds to the mathematical process of integration*)

$$V_{OUT} = \frac{1}{RC} \int_0^t V_{IN(t)} dt$$



Calculus of RC circuits response

- Response of a linear circuit with one time constant to a single-step input signal follows the equation:

$$y(t) = y(\infty) + [y(0) - y(\infty)]e^{-\frac{t}{\tau}}$$

- Time $t=t''-t'$ for what output $y(t)$ switches its value from $y(t')$ to $y(t'')$; may be calculated with equation:

$$t = t'' - t' = \tau \ln \frac{y(\infty) - y(t')}{y(\infty) - y(t'')}$$

Calculus of RC circuits response

- Response of a RC circuit to any input signal may be calculated with the Duhamel integral if the response to a single-step input is known:

$$e(t) = i(0)A(t) + \int_0^t \left. \frac{di(t)}{dt} \right|_{t=\tau} A(t - \tau) d\tau$$

- $i(t)$ – input signal
- $i(0)$ – input signal value at moment $t=0$
- $e(t)$ – circuit response (output)
- $A(t)$ – circuit response to a single-step input signal with an amplitude of 1

Alternative forms of Duhamel integral

$$e(t) = i(t)A(0) + \int_0^t \frac{dA(t)}{dt} \Big|_{t=\tau} i(t-\tau) d\tau$$

$$e(t) = i(0)A(t) + \int_0^t \frac{di(t)}{dt} \Big|_{t=t-\tau} A(\tau) d\tau$$

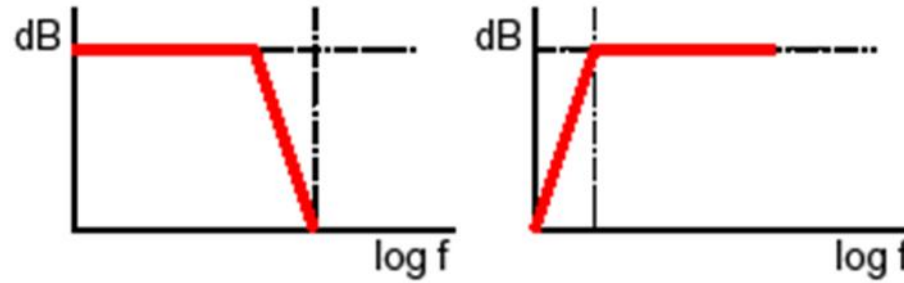
$$e(t) = i(t)A(0) + \int_0^t \frac{dA(t)}{dt} \Big|_{t=t-\tau} i(\tau) d\tau$$

Filters

- Filters are used to remove *unwanted* bandwidths from a signal
- Filter classification according to implementation:
 - Active filters include RC networks and op-amps
 - Suitable for low frequency, small signal
 - Active filters are preferred since avoid the bulk and non-linearity of inductors and can have gains greater than 0dB
 - However, active filters require a power supply
 - Passive filters consist of RCL networks
 - Simple, more suitable for frequencies above audio range, where active filters are limited by the op-amp bandwidth
 - Digital filters
 - A digital filter uses a digital processor to perform numerical calculations on sampled values of the signal. The processor may be a general-purpose computer such as a PC, or a specialized DSP (Digital Signal Processor) chip

- Filter classification according to frequency response:

- Low-pass filter
- High-pass filter



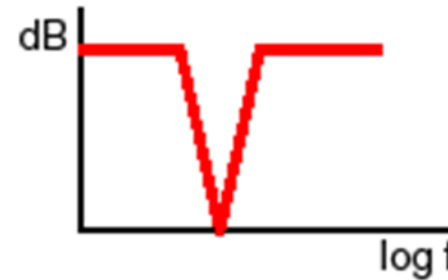
Low-pass

High-pass

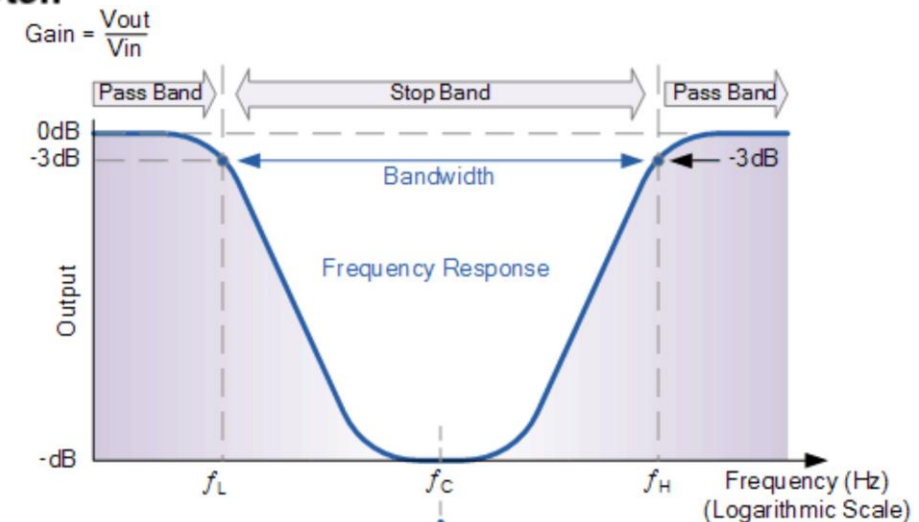
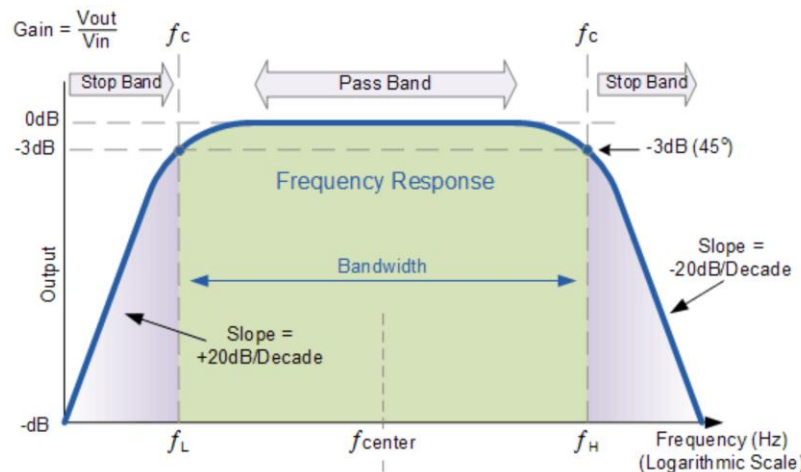
- Band-pass filter
- Band-stop (Notch) filter



Band-pass



Notch



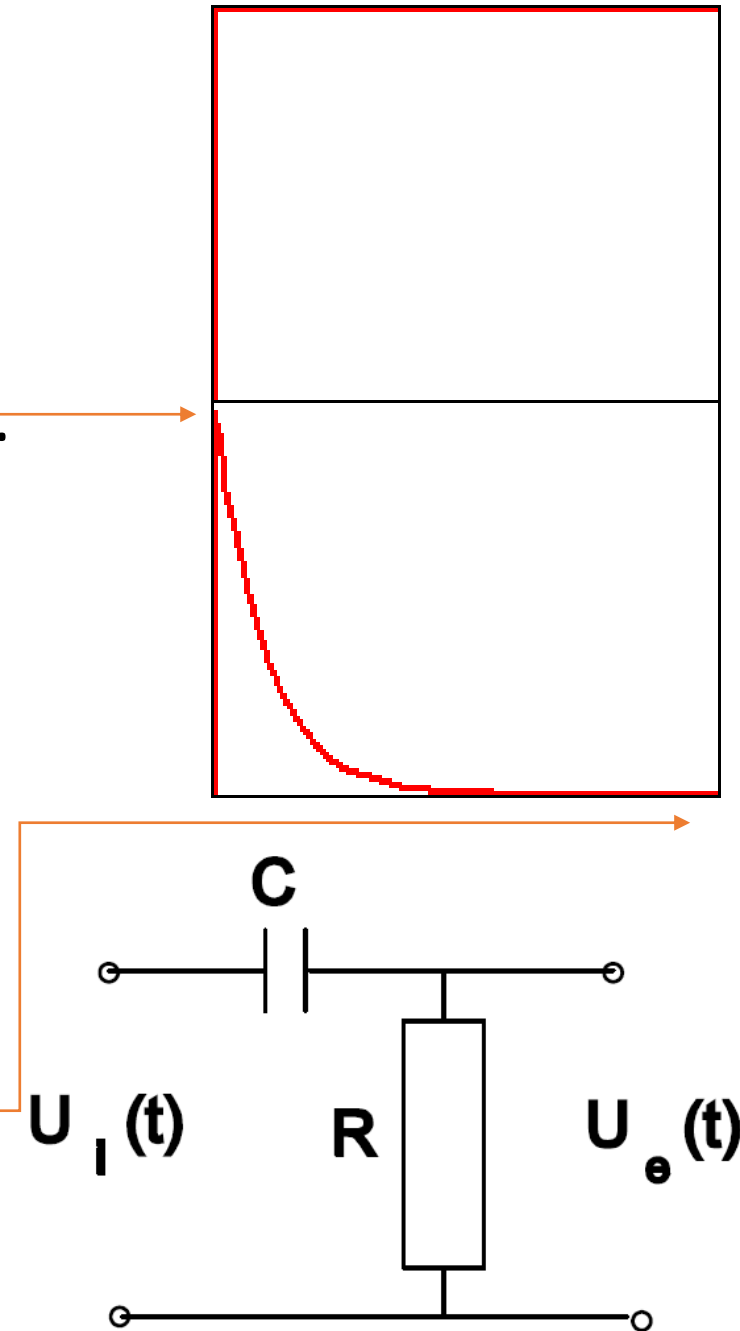
Problems

- An exponential type signal is applied at the input of a high-pass filter. Calculate and draw the circuit response.
- Duhamel integral will be used. Firstly, will be calculated the **response** of this circuit to a **single step unit** signal, e.g. the function $A(t)$:

$$u_e(t) = u_e(\infty) + [u_e(0) - u_e(\infty)]e^{-\frac{t}{\tau}}$$

$$u_e(\infty)=0; u_e(0)=1; \text{ circuit time constant } \tau = R \cdot C$$

$$u_e(t) = A(t) = e^{-t/RC}$$



The exponential input signal: $u_i(t) = U(1 - e^{-t/T})$

Applying Duhamel integral: $u_e(t) = u_i(0)A(t) + \int_0^t \frac{du_i(t)}{dt} \Big|_{t=\tau} A(t-\tau) d\tau$

With: $A(t) = e^{-t/RC}$ and $u_i(0)=0$ and:

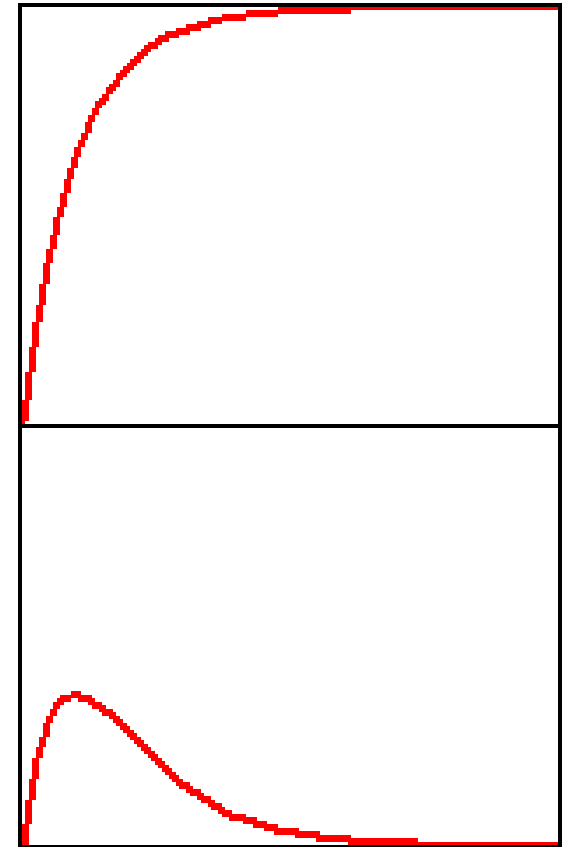
$$\frac{du_i(t)}{dt} \Big|_{t=\tau} = \frac{U}{T} e^{-\frac{\tau}{T}} \Big|_{t=\tau} = \frac{U}{T} e^{-\frac{\tau}{T}}$$

Applying the integral:

$$u_e(t) = \int_0^t \frac{U}{T} e^{-\frac{\tau}{T}} \cdot e^{-\frac{(t-\tau)}{RC}} d\tau = \frac{U}{T} e^{-\frac{t}{RC}} \int_0^t e^{\tau \frac{T-RC}{TRC}} d\tau$$

And finally:

$$u_e(t) = \frac{U}{1 - \frac{T}{RC}} \left(e^{-\frac{t}{T}} - e^{-\frac{t}{RC}} \right)$$



Proposed problems

- 1. Same problem for the low-pass filter.
- 2. Consider a high-pass filter, with components: $R=10\text{k}\Omega$, $C=100\text{nF}$. Applying at input a single step signal with amplitude of 5V, calculate the time needed for the output signal to reach the value of 2V.